



Zeitenwende: Showing in Numbers How the 2022 Russian Invasion of Ukraine Began an Arms Reshuffling Across Europe

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```
In [1]: # Google Colab instructions
#
# Runtime > Change Runtime Type > Runtime Version > 2025.07
# This cell should say you're on Python Version 3.11.

import sys
print(sys.version)
```

3.11.13 (main, Jun 4 2025, 08:57:29) [GCC 11.4.0]

1. About the Data

An extensive amount of data cleaning was done to transform the following datasets into one clean, usable spreadsheet:

- SIPRI Arms Transfer Database
- SIPRI Arms Industry Database
- SIPRI Military Expenditure Database
- Geopolitical Risk Index (Caldara & Iacoviello, 2022)
- Google Maps data (*aerial_distance_moscow_km*)

The Stockholm International Peace Research Institute (SIPRI) is a globally renowned think tank known for tracking the global arms industry. Their datasets have been used in countless peer-reviewed publications.

The SIPRI datasets obtained contained missing values. Linear interpolation was the constant method used to fill these missing values when appropriate. Several

features were created from calculations involving other features: *import_export_ratio*, *import_growth_rate*, *export_coverage_ratio*, *ME_spending_%_GDP_growth_rate*, and *export_growth_rate*. Aside from a later experiment with Gephi, all data used is broken down across the 4 years since the Russian invasion of Ukraine (2022-2025). All data is exclusive to European countries.

The cleaned dataset can be found here: https://github.com/ClaytonGreenwell/defense_spending_project/blob/main/final_submission/sipri_data1.xlsx

There are many columns in this dataset that function as a target, with an abundance of overlap:

- *import_TIV_millions*
- *import_export_ratio*
- *import_growth_rate*
- *export_coverage_ratio* (calculation relies on *import_TIV_millions*)
- *ME_spending_%_GDP*
- *ME_spending_%_GDP_lead_1_year*
- *ME_per_capita_in_USD*
- *ME_%_gov_budget*
- *ME_spending_%_GDP_growth_rate*

As an example, it's understandable that a country's arms imports (*import_TIV_millions*) will correspond directly to that country's military expenditure as a percentage of GDP (*ME_spending_%_GDP*). This overlap is true of all above-mentioned columns and is dealt with appropriately in this report.

Some of the most interesting and usable features include:

- *export_TIV_millions*
- *export_growth_rate*
- *aerial_distance_moscow_km*
- *GPR_index*
- *is_in_NATO?*

- *top_supplier_country*
- *is_US_top_supplier_country?*
- *companies_in_sipri_top_100*

Many countries in *companies_in_sipri_top_100* contain fractional values. This is due to joint-EU defense companies that spread operations across multiple EU countries without assigning ownership to a single state.

Some notes on the choices made for visualizations throughout this report:

- The definition of “Europe” includes all countries that are members of the Council of Europe.
 - o Russia is included, though it was expelled from the Council in 2022.
 - o Belarus and Kosovo were included, despite not being members of the Council.
- Five “microstates” were excluded from all visualizations: Andorra, Liechtenstein, Monaco, San Marino, and Vatican City.
- Malta was included. Malta was not classified as a “microstate” despite its size and population for 3 reasons:
 - o It is a European Union member.
 - o Unlike the other 5 microstates, Malta doesn’t have a dramatic diplomatic dependence on other states.
 - o Malta’s location in the Mediterranean Sea is strategically important from a military defense perspective.
- Various disputed states were excluded: Northern Cyprus, Transnistria, Abkhazia, and South Ossetia.
- Article 41(2) of the Treaty on European Union prohibits the regular EU budget from being used for military defense purposes. Some EU states have contributed to a workaround agency called The European Peace Facility. The EPF purchases arms as a unified body. Many of these arms are given to Ukraine or are used to bolster a collective NATO/EU defense network (Cepparulo & Pasimeni, 2024). As this is a

collective agency that operates on the fringe of the EU, it's important to understand that imports and exports from this agency would not appear in import/export visualizations unless an actual country (like Ukraine) is receiving those arms as an import.

```
In [2]: # We'll need pycaret for creating our machine learning models later on. We'll
# need to install pycaret before we import our data, as we'll otherwise lose
# our data when restarting the session.
#
# If running this cell in your own environment, note that removing
# "> /dev/null 2>&1" will show the extended output and errors. Removing this
# will also conclude the output by manually prompting you to restart the
# session (if using Google Colab).

!pip install pycaret openpyxl > /dev/null 2>&1
```

```
In [1]: # Due to installing pycaret, before running this cell:
# Runtime > Restart Session

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: df = pd.read_excel("sipri_data1.xlsx")
```

2. Tableau Visualization I: Arms Exports

Europe's Engine

Figure 1:

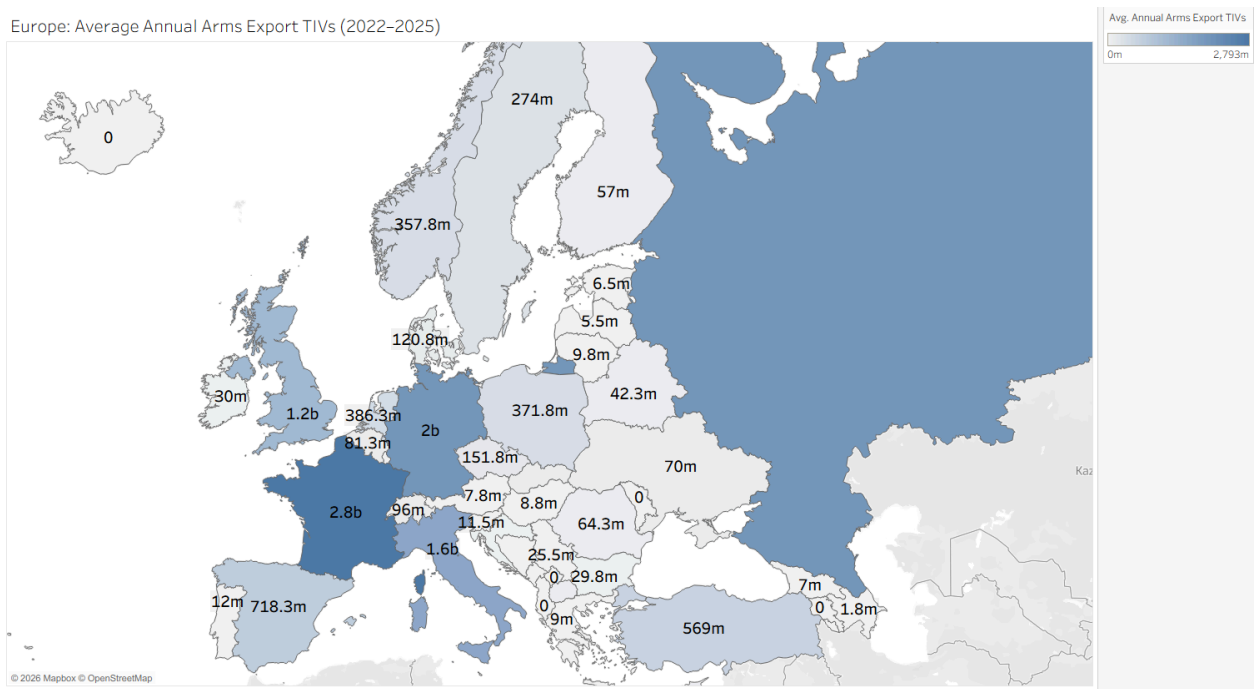
<https://public.tableau.com/app/profile/clayton.greenwell/viz/EuropeAverageAnnualArmsExportTIVs20222025/Sheet1>

The above link is interactive. Here is a static image of Figure 1:

```
In [3]: from IPython.display import Image, display

display(Image(filename="export_tableau.png", width=1100))
```

Europe: Average Annual Arms Export TIVs (2022–2025)



Some notes on this visualization:

- This visualization averages annual arms export data from 2022 – 2025. (The wider Russian ground invasion of Ukraine began in 2022).
- The unit of measurement used in this visualization is Trend Indicator Values, commonly known as TIV units. TIV units are the gold standard for arms measurements because they help normalize what would otherwise be a very difficult calculation. (Keep in mind that using euros or dollars sold wouldn't be practical in a long-term study, as there were long periods during the Cold War that the US and USSR gave away weapons to other countries for free). As a measurement tool, TIV units standardize production costs by evaluating raw capability and resources required to build hardware (SIPRI, n.d.).

Germany, France, Italy, and the United Kingdom are strong European engine economies that have the capabilities to create and export an incredible number of arms. These 4 countries together have an average annual output of 7.47 billion TIV units since the start of the war. SIPRI compiles a yearly list of the Top 100 Global Defense Companies. Germany, France, Italy, and the UK have collectively comprised roughly 20% of the companies listed since the outbreak of war in 2022 (SIPRI, n.d.).

Despite tough foreign sanctions and the ongoing domestic needs of immediate warfare, Russia has managed to export an average of 2 billion TIV units a year since the start of the war. Arms sales to countries like India have helped to bolster

Russia's struggling economy (SIPRI, n.d.). On the world market, many of Russia's most sought-after weapons rely in some small part on small components manufactured in the US, the UK, and other Western countries that are attempting to enforce sanctions and export control restrictions (George & Sandler, 2022). Export control restrictions are meant to throttle Russia's supply of these dual-use components. Russia's inability to access advanced computer chip technology is a critical chokepoint affecting Russia's arms industry (George & Sandler, 2022).

3. Tableau Visualization II: Arms Imports

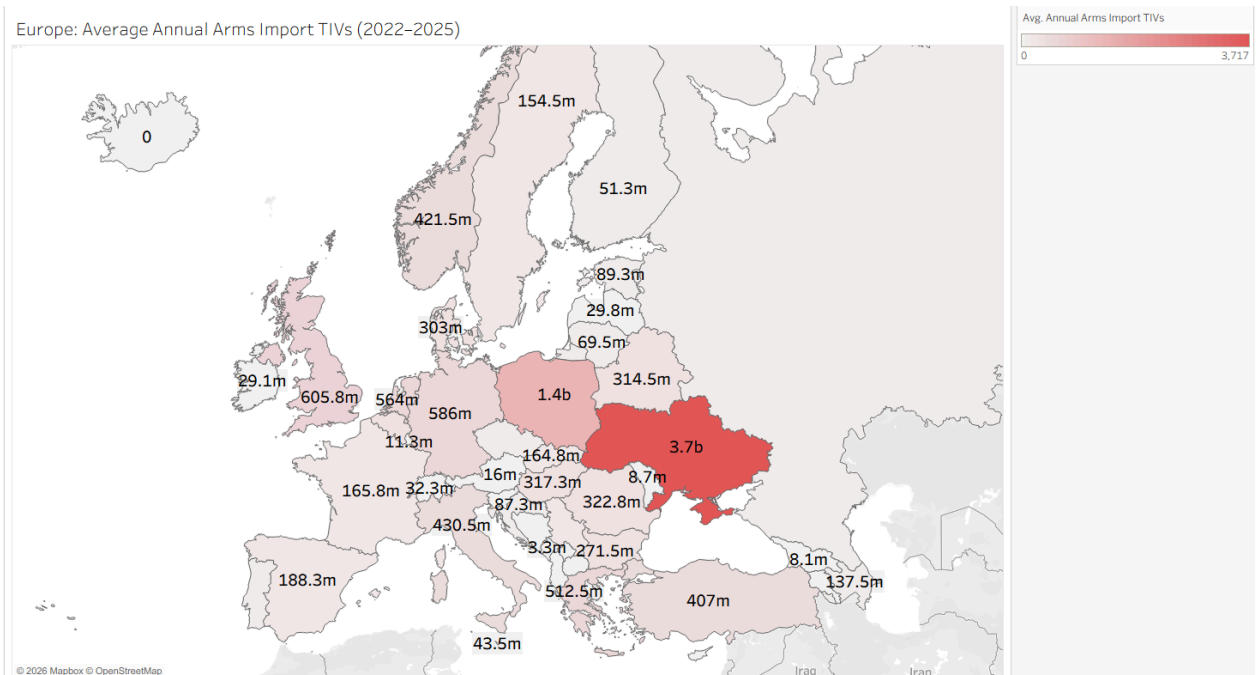
Russia Anxiety in Poland

Figure 2:

<https://public.tableau.com/app/profile/clayton.greenwell/viz/EuropeAverageAnnualArmsImportTIVs2022-2025/Sheet1>

The above link is interactive. Here is a static image of Figure 2:

```
In [4]: display(Image(filename="import_tableau.png", width=1100))
```



Ukraine's massive annual import of 3.72 billion TIV units worth of arms sits as an outlier, massively tipping the color scale and leaving other countries unsaturated.

Despite this outlier, important information can be gleaned by country shading that isn't as immediately stark. Poland's annual average import of 1.38 billion TIV units


```

filename="top_arms_supplier_relationships_before_invasion_2018_2021.png"
width=1100
)
)

```

Top Arms Supplier Relationships Before Invasion (2018-2021)

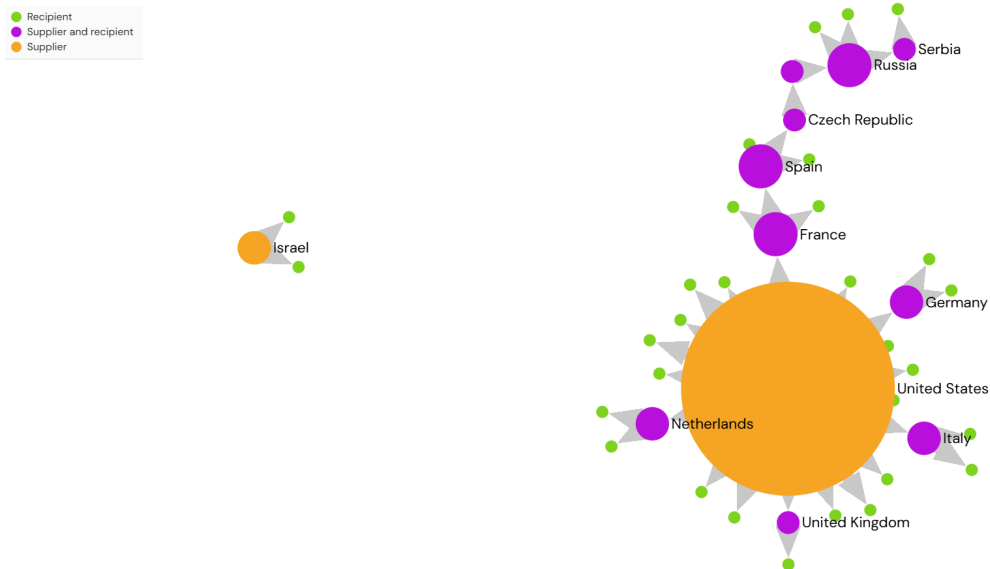


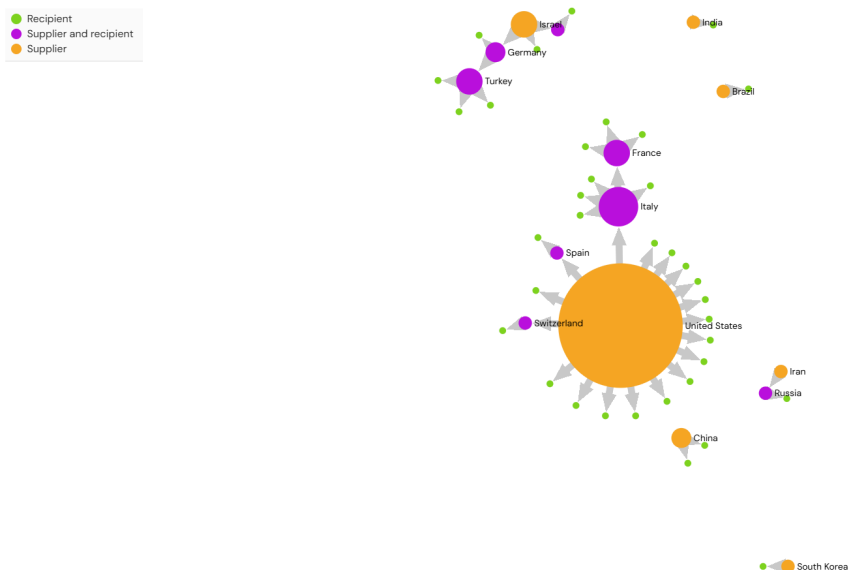
Figure 4:

```

In [6]: display(
    Image(
        filename="top_arms_supplier_relationships_after_invasion_2022_2025.png"
        width=1100
    )
)

```

Top Arms Supplier Relationships After Invasion (2022-2025)



Gephi is a powerful visualization tool for building network models. Perfect for visualizing supplier-recipient relationships, it gives us unique insight into how Russia's 2022 invasion of Ukraine disrupted a surprisingly well-connected European arms network.

With the invasion beginning in 2022, we have 4 full years of post-invasion data (2022-2025). Using these 4 years, we look at the total TIV units imported by every European country to determine its top supplier (SIPRI, n.d.). This metric relies on *plurality*. For example, a country could have a very diverse supplier profile with their top supplier only making up 30% of their imports. But that top supplier would still be the most important defense relationship—and it would be reflected in this metric. Figure 3 shows the 4 years leading up to the war (2018-2021) to give us an idea of what the supplier-recipient relationships looked like before the disruption.

In Figure 3, we see a remarkable amount of connectedness amongst nearly all European countries and their suppliers. The United States stands out as a giant top supplier for the most countries at 19. Even *Russia* is connected in this arms supply network, although for unusual reasons. To our surprise, Russia's top arms supplier before the outbreak of war was in fact *Ukraine*. This is due to a concept called *dual-use technology*, where a technology sold for civilian purposes can be repurposed for military use. SIPRI's 2018 data shows TIV unit transfers from Ukraine to Russia (SIPRI, n.d.). Though supply contracts were terminated between Russia and Ukraine in 2014, transferred parts (Antonov An-148-100E transport jet parts) assembled in Russia had licensing tied to a defense company based in Ukraine, with the transfer not being recorded by SIPRI until 2018 (UNIAN, 2019). Evidence is unclear as to whether Russia has used these transport jets for military purposes since the start of the invasion, but the example stands as a reminder that dual-use technology creates an incredible amount of overlap between the aerospace and defense industries.

Figure 4 shows this same European arms supply network become incredibly fragmented after the outbreak of the full-scale invasion. We see the existence of 7 unconnected networks: *the United States, Israel, India, Brazil, Iran, China, and South Korea*. The disparity in connectedness between the pre-war Figure 3 and the post-invasion Figure 4 shows us a European arms supply network that has splintered, with governments realigning their priorities and alliances.

Figure 3's pre-war arms network's surprising connectedness is not coincidental. Students of history may remember the First World War as a confounding conflict with a complicated list of reasons for its beginning. But history teachers hold on to a single foundational axiom that best sums up the powder keg that began the Great War: The system of alliances that had kept Europe at peace had broken

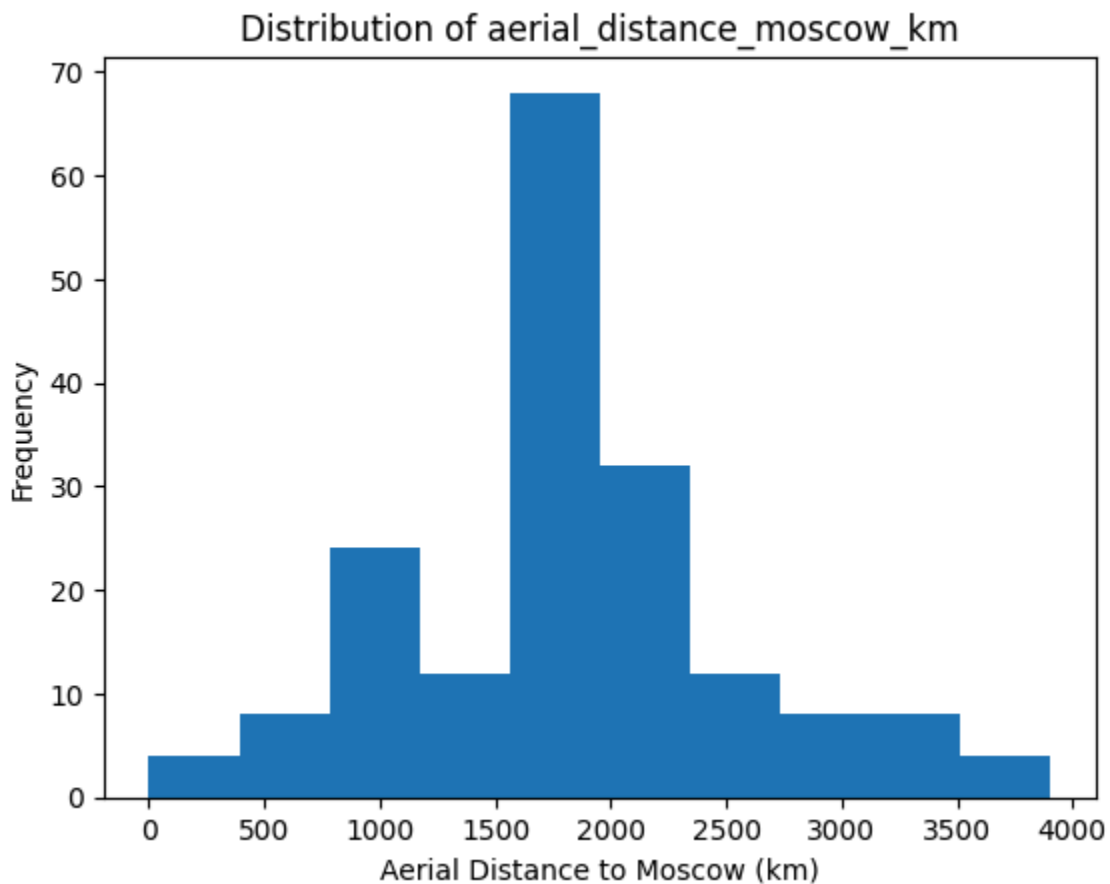
down. We can view the remarkable connectedness of the 2018-2021 European arms network modeled in Figure 3 as an individual system of alliances exclusive to weaponry and defense. It cannot be inferred that the fragmentation that is reflected in the post-invasion Figure 4 will begin a global conflict like the First World War. But it is a grave illustration of the seriousness and significance of this current conflict: Europe's first large, conventional land war since the World War period.

5. Histograms, Heatmap, and Pearson Correlation Coefficients

Digging into the coding section, we start exploring our data with some simple statistics and visualizations.

```
In [7]: # Our only feature that follows a Normal distribution
```

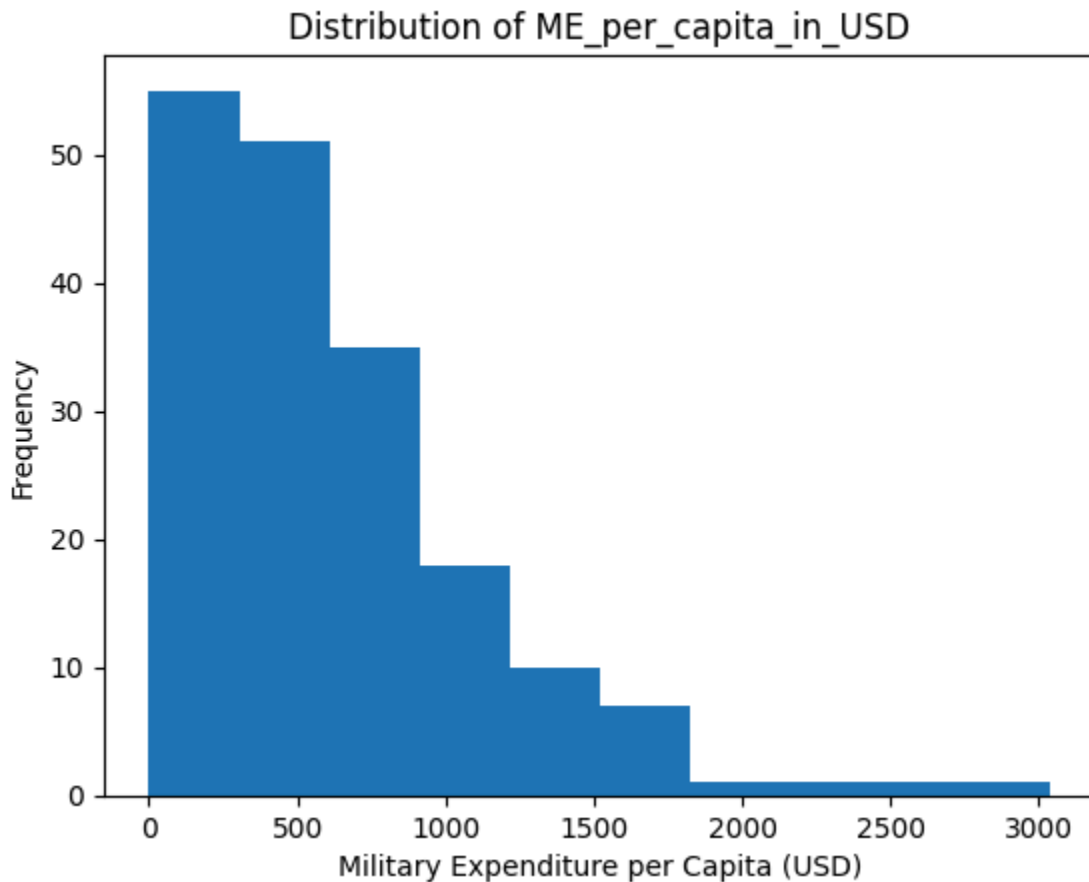
```
plt.hist(df["aerial_distance_moscow_km"])
plt.title("Distribution of aerial_distance_moscow_km")
plt.xlabel("Aerial Distance to Moscow (km)")
plt.ylabel("Frequency")
plt.show()
```



```
In [8]: # An example of a feature with a skewed distribution.
```

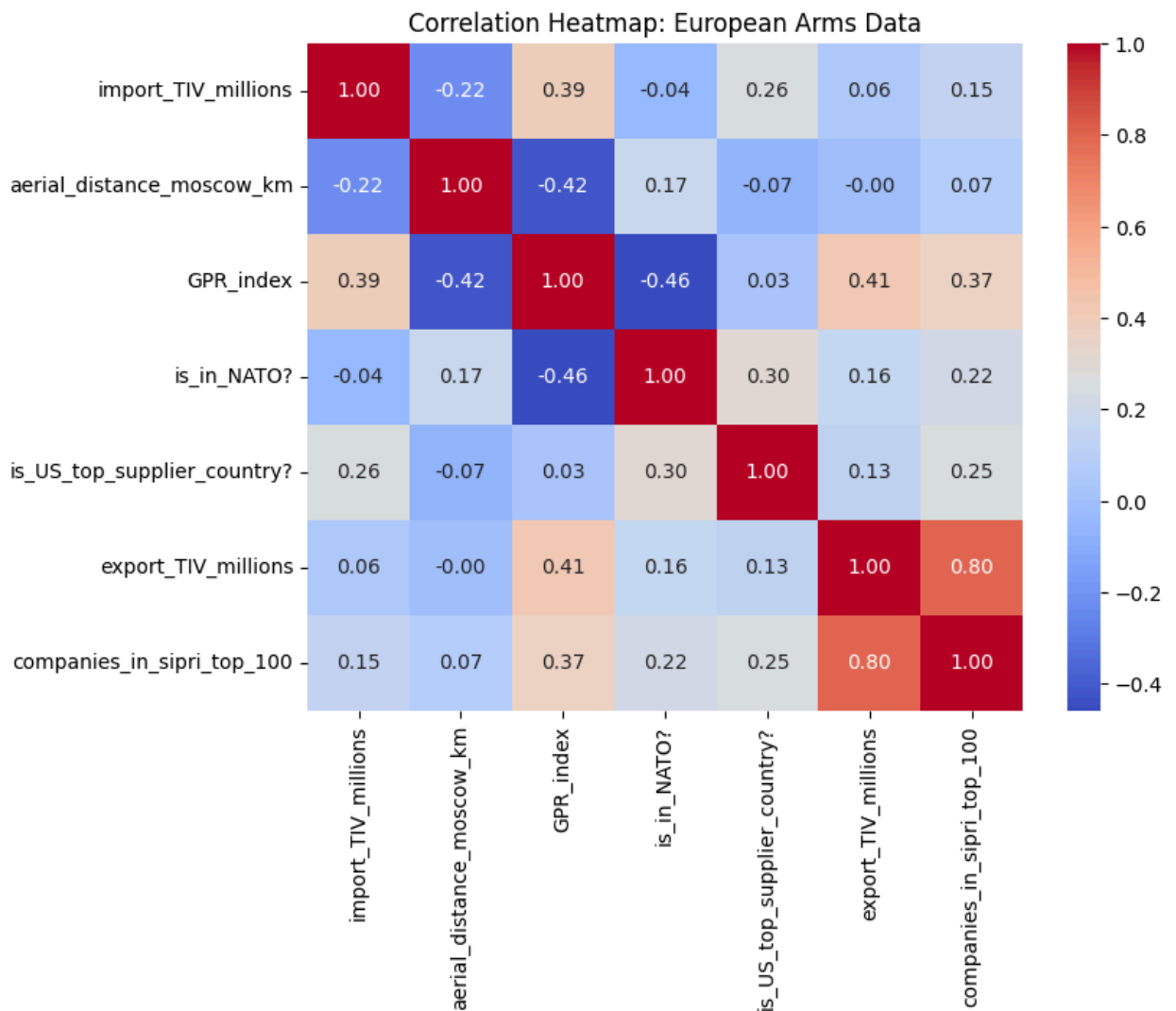
```
# Shows us that a lot of European countries spend very little per capita.
```

```
plt.hist(df["ME_per_capita_in_USD"])  
plt.title("Distribution of ME_per_capita_in_USD")  
plt.xlabel("Military Expenditure per Capita (USD)")  
plt.ylabel("Frequency")  
plt.show()
```



```
In [9]: # Heatmap of our features
```

```
features = [  
    'import_TIV_millions',  
    'aerial_distance_moscow_km',  
    'GPR_index',  
    'is_in_NATO?',  
    'is_US_top_supplier_country?',  
    'export_TIV_millions',  
    'companies_in_sipri_top_100'  
]  
  
heatmap_df = df[features].corr()  
  
plt.figure(figsize=(8, 6))  
sns.heatmap(heatmap_df, annot=True, cmap='coolwarm', fmt=".2f")  
plt.title('Correlation Heatmap: European Arms Data')  
plt.show()
```



```
In [10]: # Pearson Correlation Coefficients
```

```
# Filter down to only the year 2023, and only NATO members.
NATO_2023 = df[(df["year"] == 2023) & (df["is_in_NATO?"] == 1)]

distance_vs_ME = (
    NATO_2023["aerial_distance_moscow_km"]
    .corr(NATO_2023["ME_spending_%_GDP_lead_1_year"])
)

GPR_vs_top100 = (
    NATO_2023["GPR_index"]
    .corr(NATO_2023["companies_in_sipri_top_100"])
)

corr_table = pd.DataFrame({
    "Description": [
        "Distance From Moscow vs. ME % GDP (NATO, 2024)",
        # The reason that the label says (NATO, 2024) is that we're using
        # ME_spending_%_GDP_lead_1_year in the correlation.
        "GPR Index vs. Top 100 Defense Companies (NATO, 2023)",
    ]
})
```

```

    ],
    "Pearson Correlation": [
        distance_vs_ME,
        GPR_vs_top100
    ]
})

print(corr_table)

```

	Description	Pearson Correlation
0	Distance From Moscow vs. ME % GDP (NATO, 2024)	-0.672786
1	GPR Index vs. Top 100 Defense Companies (NATO,...	0.881429

```

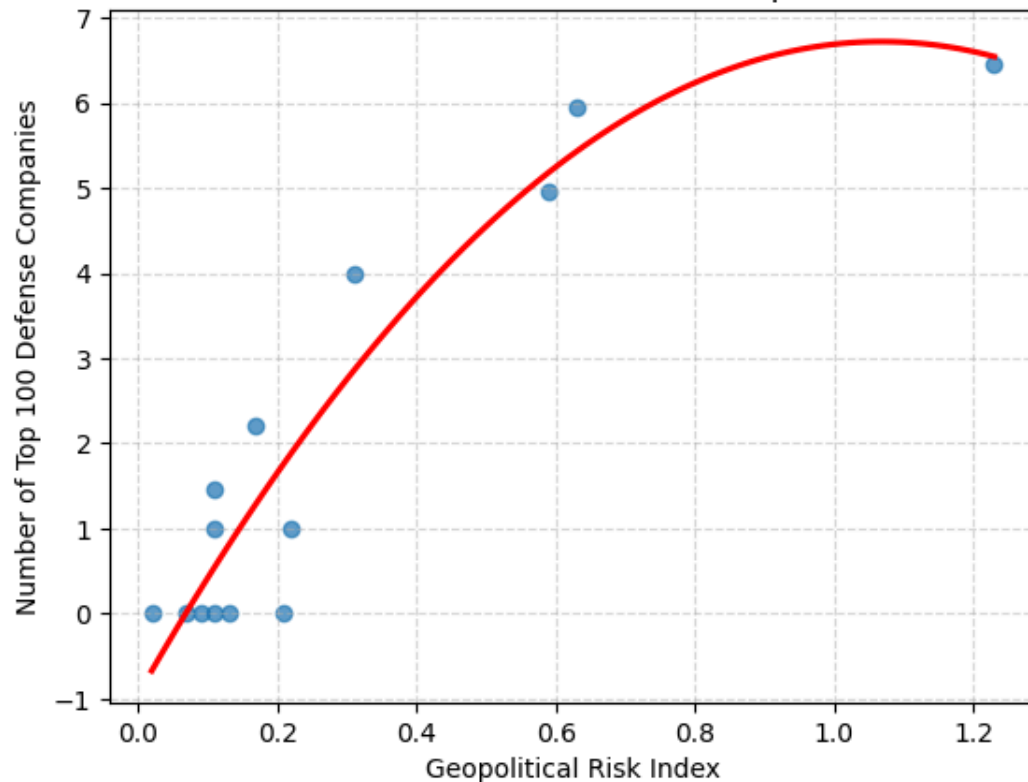
In [11]: # We're going to plot the better of the two correlations.
         # This will be polynomial regression.

sns.regplot(
    data=NATO_2023,
    x="GPR_index",
    y="companies_in_sipri_top_100",
    order=2, # This will make it look like 2nd-degree polynomial regression
    ci=None,
    scatter_kws={"alpha": 0.7},
    line_kws={"color": "red"}
)

plt.xlabel("Geopolitical Risk Index")
plt.ylabel("Number of Top 100 Defense Companies")
plt.title("NATO Members (2023): GPR Index vs. Number of Top 100 Defense Compar")
plt.grid(True, linestyle="--", alpha=0.5)
plt.show()

```

NATO Members (2023): GPR Index vs. Number of Top 100 Defense Companies



We explore this here instead of in the regression section because we have enough data points to make a neat visualization, but not enough for the more complex regression modeling coming later. GPR Index as a feature has lots of missingness, as not all European countries were given values.

6. Regression Modeling: The Year That Rebalanced Europe

How 2023 Became a Powerful Predictor of Military Spending

```
In [12]: from pycaret.regression import *  
  
# We'll build the first model pipeline.  
  
regression_pipeline = setup(  
    data=df,  
    target='ME_spending_%_GDP',  
    categorical_features=['year'],  
    ignore_features=[  
        'GPR_index',  
        'companies_in_sipri_top_100',  
        'is_in_NATO?',  
        'export_coverage_ratio',  
        'export_TIV_millions',
```

```
        'is_US_top_supplier_country?',
        'top_supplier_country',
        'import_TIV_millions',
        'country',
        'ME_spending_%_GDP_lead_1_year',
        'ME_per_capita_in_USD',
        'ME_%_gov_budget',
        'ME_spending_%_GDP_growth_rate',
        'import_growth_rate',
        'import_export_ratio',
        'export_growth_rate',
    ],
    numeric_imputation='median',
    #categorical_imputation='constant',
    normalize=True,
    session_id=42,
    train_size=0.8,
    fold = 5
)
```

	Description	Value
0	Session id	42
1	Target	ME_spending_%_GDP
2	Target type	Regression
3	Original data shape	(180, 19)
4	Transformed data shape	(180, 6)
5	Transformed train set shape	(144, 6)
6	Transformed test set shape	(36, 6)
7	Ignore features	16
8	Numeric features	1
9	Categorical features	1
10	Rows with missing values	70.0%
11	Preprocess	True
12	Imputation type	simple
13	Numeric imputation	median
14	Categorical imputation	mode
15	Maximum one-hot encoding	25
16	Encoding method	None
17	Normalize	True
18	Normalize method	zscore
19	Fold Generator	KFold
20	Fold Number	5
21	CPU Jobs	-1
22	Use GPU	False
23	Log Experiment	False
24	Experiment Name	reg-default-name
25	USI	d789

```
In [13]: best_model = compare_models()
```


	Model	MAE	MSE	RMSE	R2	RMSLE	MAPE	TT (Sec)
xgboost	Extreme Gradient Boosting	0.0048	0.0001	0.0084	0.8164	0.0074	0.2512	0.1520
huber	Huber Regressor	0.0116	0.0017	0.0320	0.0284	0.0278	0.5901	0.0660
lasso	Lasso Regression	0.0147	0.0018	0.0342	-0.3794	0.0300	1.0405	0.0880
dummy	Dummy Regressor	0.0147	0.0018	0.0342	-0.3794	0.0300	1.0405	0.0940
llar	Lasso Least Angle Regression	0.0147	0.0018	0.0342	-0.3794	0.0300	1.0405	0.0560
en	Elastic Net	0.0147	0.0018	0.0342	-0.3794	0.0300	1.0405	0.0540
omp	Orthogonal Matching Pursuit	0.0157	0.0017	0.0344	-0.7290	0.0293	0.8826	0.0600
br	Bayesian Ridge	0.0153	0.0018	0.0344	-0.7490	0.0297	0.9028	0.0540
et	Extra Trees Regressor	0.0154	0.0021	0.0364	-0.8341	0.0319	0.6916	0.1500
knn	K Neighbors Regressor	0.0152	0.0019	0.0370	-1.0720	0.0325	0.7914	0.0640
ridge	Ridge Regression	0.0169	0.0018	0.0360	-1.3943	0.0307	0.9553	0.0760
lr	Linear Regression	0.0169	0.0018	0.0360	-1.4217	0.0307	0.9582	1.8120
lar	Least Angle Regression	0.0169	0.0018	0.0360	-1.4217	0.0307	0.9582	0.0580
lightgbm	Light Gradient Boosting Machine	0.0158	0.0019	0.0370	-1.5086	0.0326	0.9289	0.2660
gbr	Gradient Boosting Regressor	0.0073	0.0007	0.0216	-1.5598	0.0182	0.3474	0.1100
rf	Random Forest Regressor	0.0100	0.0009	0.0271	-2.2915	0.0234	0.4610	0.1940
ada	AdaBoost Regressor	0.0120	0.0015	0.0304	-3.4243	0.0264	0.7545	0.1000
dt	Decision Tree Regressor	0.0098	0.0018	0.0312	-4.4893	0.0267	0.4216	0.0560

	Model	MAE	MSE	RMSE	R2	RMSLE	MAPE	TT (Sec)
par	Passive Aggressive Regressor	0.0513	0.0049	0.0689	-24.6622	0.0593	3.4693	0.0540

```
In [14]: xgboost_model = create_model('xgboost')
```

	MAE	MSE	RMSE	R2	RMSLE	MAPE
Fold						
0	0.0029	0.0000	0.0040	0.7350	0.0040	0.2545
1	0.0039	0.0000	0.0050	0.7872	0.0049	0.2475
2	0.0058	0.0002	0.0130	0.9650	0.0106	0.1597
3	0.0052	0.0001	0.0084	0.6333	0.0081	0.3630
4	0.0061	0.0001	0.0118	0.9616	0.0094	0.2313
Mean	0.0048	0.0001	0.0084	0.8164	0.0074	0.2512
Std	0.0012	0.0001	0.0036	0.1298	0.0026	0.0653

```
In [15]: tuned_xgboost = tune_model(xgboost_model, choose_better=True)
```

	MAE	MSE	RMSE	R2	RMSLE	MAPE
Fold						
0	0.0031	0.0000	0.0035	0.8035	0.0034	0.2674
1	0.0038	0.0000	0.0045	0.8242	0.0044	0.2701
2	0.0131	0.0025	0.0501	0.4837	0.0400	0.2073
3	0.0045	0.0001	0.0078	0.6831	0.0075	0.2732
4	0.0037	0.0000	0.0051	0.9928	0.0050	0.2112
Mean	0.0056	0.0005	0.0142	0.7575	0.0121	0.2458
Std	0.0037	0.0010	0.0180	0.1688	0.0140	0.0300

Fitting 5 folds for each of 10 candidates, totalling 50 fits
Original model was better than the tuned model, hence it will be returned. NOT
E: The display metrics are for the tuned model (not the original one).

```
In [16]: from pycaret.regression import get_config
```

```
X_train = get_config('X_train')
X_train.head()
```

Out[16]:

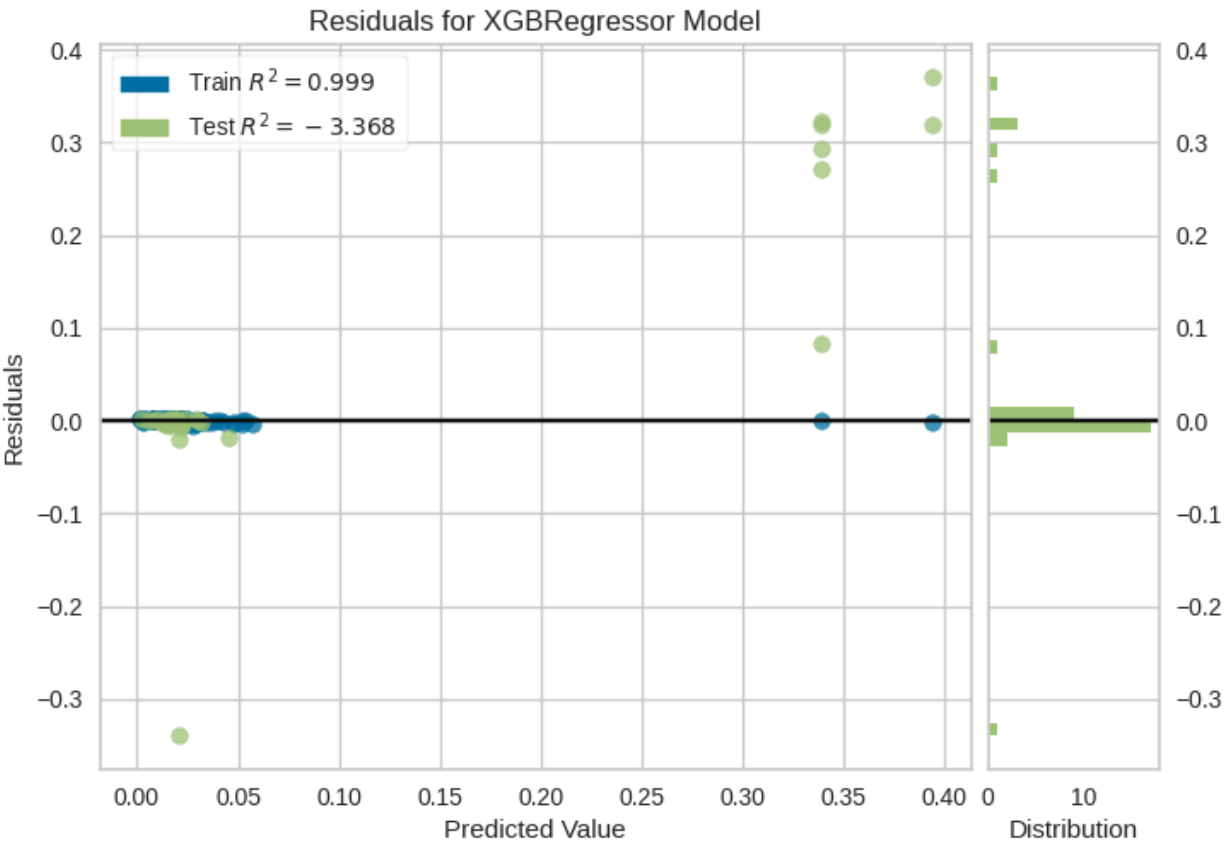
	year	aerial_distance_moscow_km
117	2023	2146.979980
65	2023	1608.829956
67	2025	1608.829956
31	2025	1777.410034
12	2022	1927.380005

```
In [17]: X_train_process = get_config('X_train_transformed')
X_train_process.head()
```

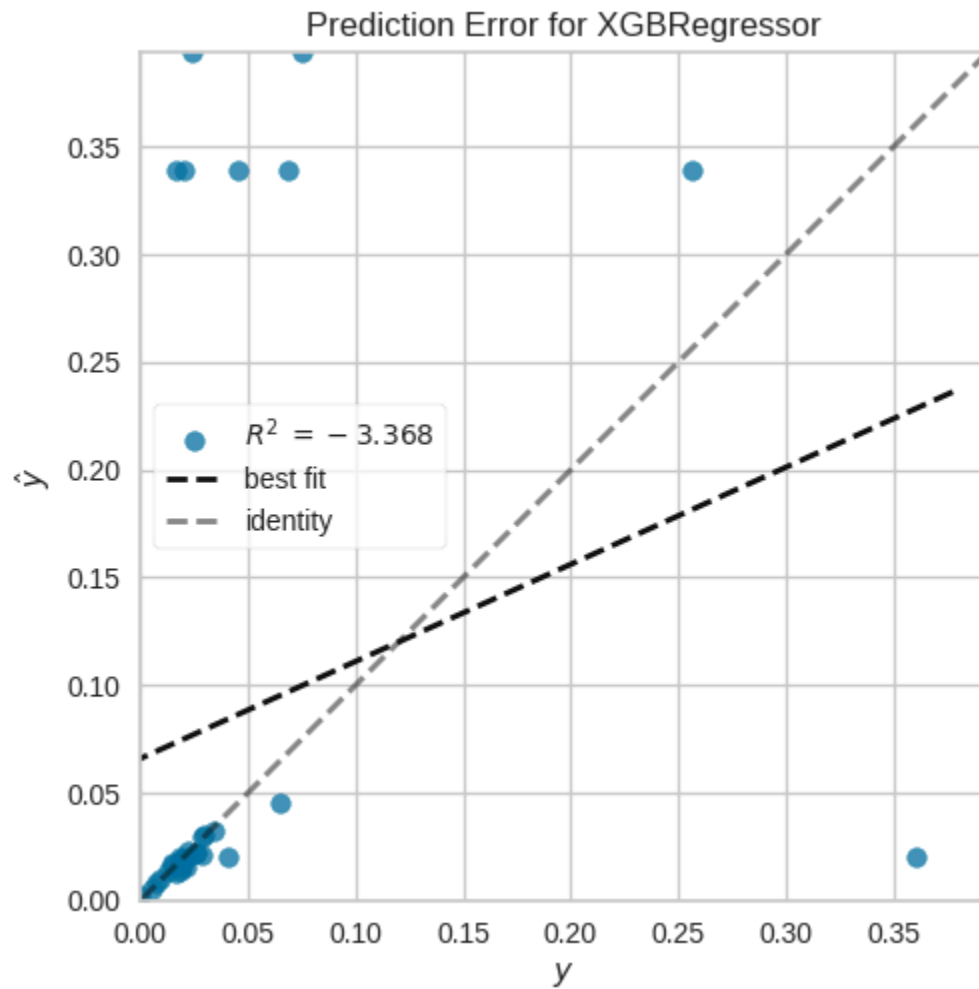
Out[17]:

	year_2023.0	year_2025.0	year_2022.0	year_2024.0	aerial_distance_moscow_km
117	1.834022	-0.609449	-0.566658	-0.588043	0.5
65	1.834022	-0.609449	-0.566658	-0.588043	-0.4
67	-0.545250	1.640825	-0.566658	-0.588043	-0.4
31	-0.545250	1.640825	-0.566658	-0.588043	-0.2
12	-0.545250	-0.609449	1.764734	-0.588043	0.0

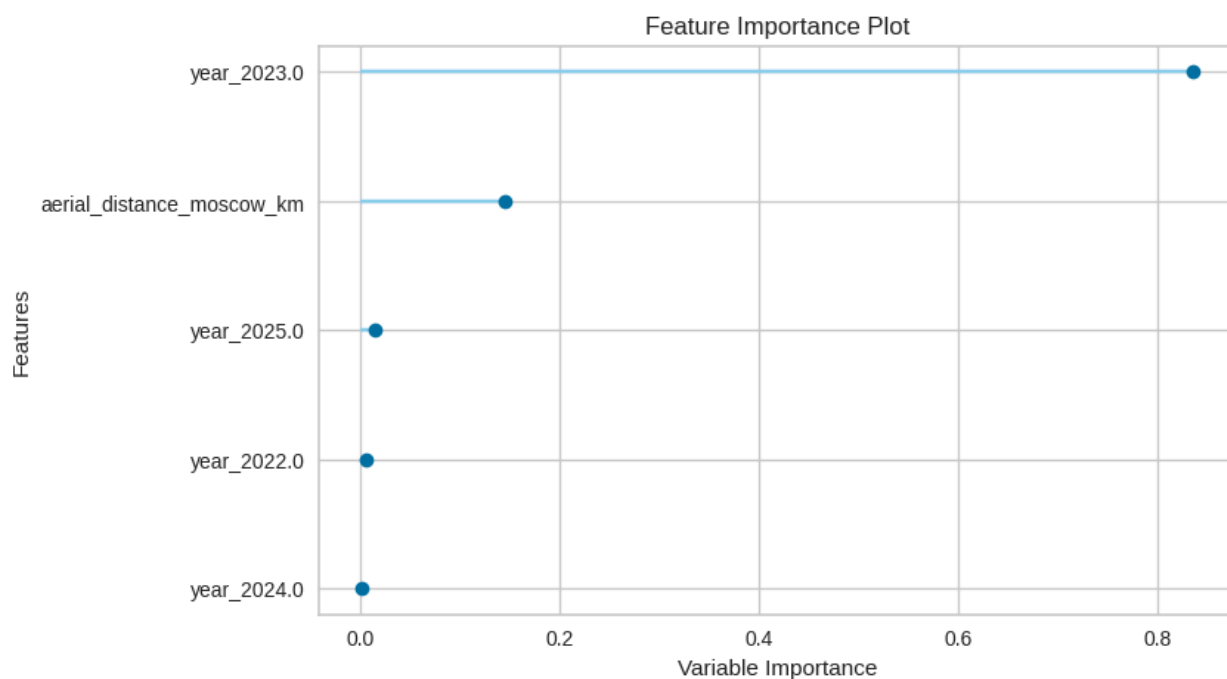
```
In [18]: plot_model(tuned_xgboost, plot='residuals')
```



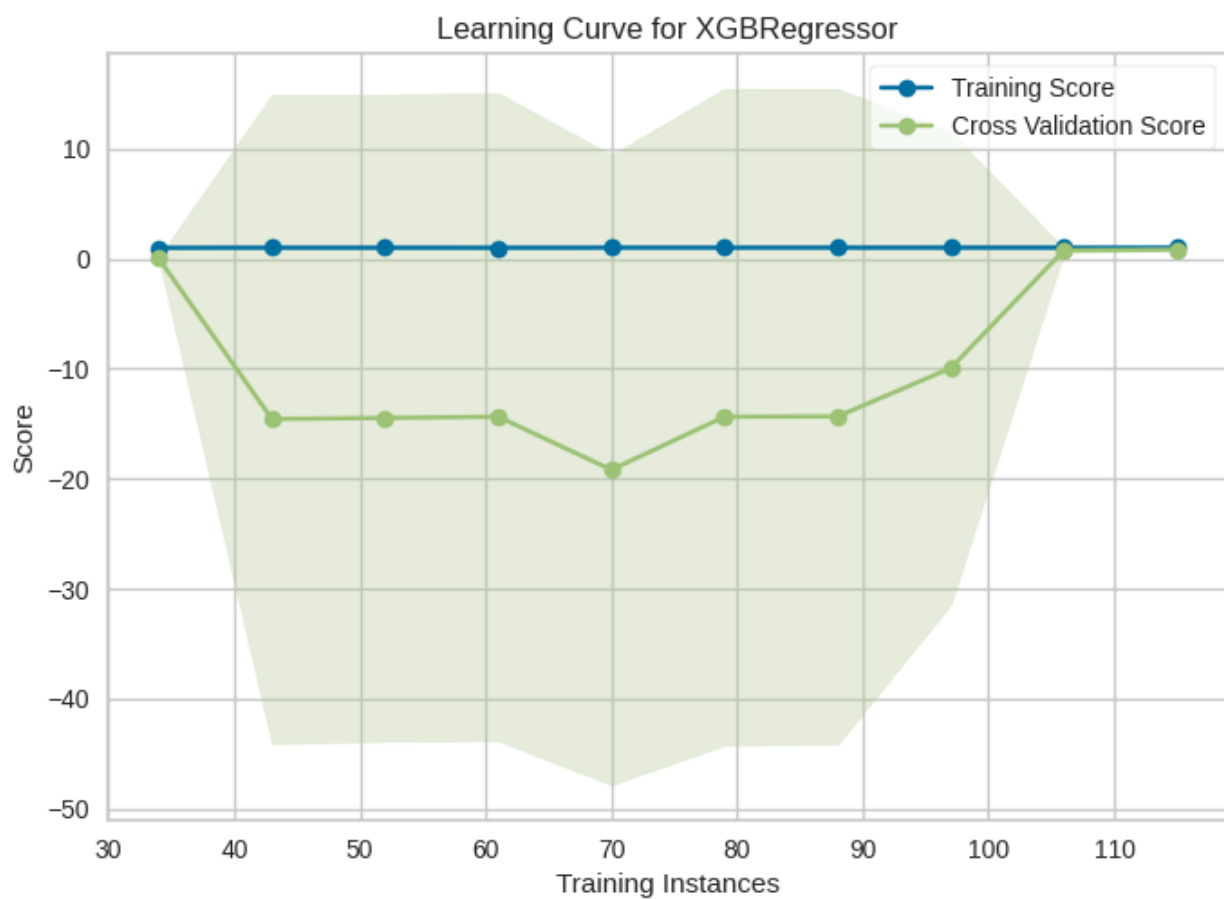
```
In [19]: plot_model(tuned_xgboost, plot='error')
```



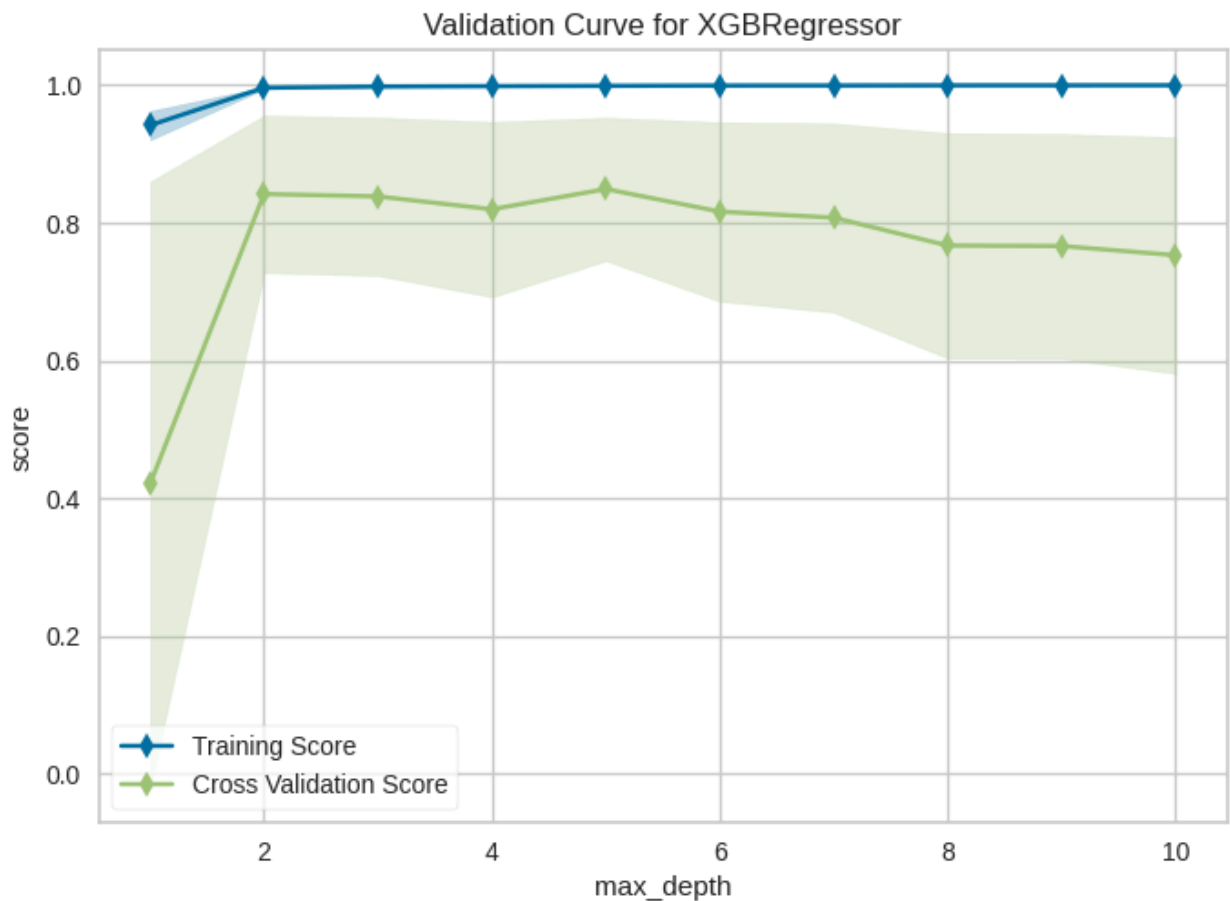
```
In [20]: plot_model(tuned_xgboost, plot='feature')
```



```
In [21]: plot_model(tuned_xgboost, plot="learning")
```



```
In [22]: plot_model(tuned_xgboost, plot="vc")
```



Following the February 2022 invasion of Ukraine, European governments braced for a rebalancing of the world order. German Chancellor Olaf Scholz deemed this moment to be Europe's *Zeitenwende*—a historic turning point (Cepparulo & Pasimeni, 2024).

As governments create central budgets in the year prior based on geopolitical risks encountered recently, it can be derived that a massive and unpredicted reshuffling of risk in 2022 would be first noticed in most European government spending patterns in calendar year 2023. We see this global reshuffling play out precisely in our regression model. When using military expenditure as a share of GDP as our target, we are left with 2 features that are of the most importance: data from calendar year 2023 (> 80%) and the aerial distance that a European capital sits from Moscow (~15%). With 4 years of war sampled in this dataset, (2022 through 2025,) it would be understandable that the year 2023 would bear the most importance, as it is the first year in which central budgets across Europe were given a chance to plan for and redirect capital toward growing uncertainty in the East.

To explain why our additional feature *aerial_distance_moscow_km* played significant importance, we need to explain the inspiration behind its inclusion in

the dataset. Knezović and Tkalec (2025) posed an observation: For every 100 kilometers that a NATO member's capital is located farther from Moscow, their average yearly military expenditure as a percentage of GDP has an associated decrease of 0.02 percentage points. Knezović and Tkalec (2025) found that Western NATO members, protected by geography, distance, and buffer states, were much less likely to see the rise of Russian aggression as an immediate threat demanding significant defense budget changes.

Extreme Gradient Boosting proved to be the best performing regression model, with a MAE of 0.0048, MSE of 0.0001, a RMSE of 0.0084, and an R-squared of 0.82. It is important to note that the target, *ME_spending_%_GDP*, is stored as a proportion: 0.03 meaning 3% of GDP. Therefore, an MAE of 0.0048 means approximately 0.48 percentage points, and not 0.0048 percentage points.

Extreme Gradient Boosting was uniquely positioned to excel where other regression techniques failed, as this was a fairly small dataset, (only 180 records,) and Extreme Gradient Boosting is known to handle interactions well. Boosted tree structured models are better at modeling nonlinear relationships, as well as working with the necessary inclusion of outliers, (Ukraine and Russia,) whereas other regression models could easily be pulled astray.

7. Classification Modeling I: United States as Top Supplier Country

```
In [23]: from pycaret.classification import *
```

```
In [24]: classification_pipeline = setup(
    data=df,
    target='is_US_top_supplier_country?',
    session_id=42,
    normalize=True,
    ignore_features=[
        'country',
        'top_supplier_country',
        'year',
        'ME_spending_%_GDP_lead_1_year'
    ],
    numeric_imputation='mean',
    categorical_imputation='mode',
    fix_imbalance=True,
    fold=5
)
```

	Description	Value
0	Session id	42
1	Target is_US_top_supplier_country?	
2	Target type	Binary
3	Original data shape	(180, 19)
4	Transformed data shape	(221, 15)
5	Transformed train set shape	(166, 15)
6	Transformed test set shape	(55, 15)
7	Ignore features	4
8	Numeric features	14
9	Rows with missing values	70.0%
10	Preprocess	True
11	Imputation type	simple
12	Numeric imputation	mean
13	Categorical imputation	mode
14	Fix imbalance	True
15	Fix imbalance method	SMOTE
16	Normalize	True
17	Normalize method	zscore
18	Fold Generator	StratifiedKFold
19	Fold Number	5
20	CPU Jobs	-1
21	Use GPU	False
22	Log Experiment	False
23	Experiment Name	clf-default-name
24	USI	dd74

```
In [25]: # Compare all models
         best_model1 = compare_models()
```


	Model	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC	TF (S)
gbc	Gradient Boosting Classifier	0.7520	0.7863	0.5778	0.6460	0.6055	0.4273	0.4317	0.0
rf	Random Forest Classifier	0.7440	0.7969	0.5306	0.6514	0.5733	0.3981	0.4095	0.0
dt	Decision Tree Classifier	0.7120	0.6676	0.5278	0.5641	0.5374	0.3330	0.3372	0.0
lr	Logistic Regression	0.7040	0.7032	0.7222	0.5624	0.6192	0.3897	0.4101	0.0
qda	Quadratic Discriminant Analysis	0.7040	0.7040	0.5500	0.5811	0.5480	0.3338	0.3441	0.0
lda	Linear Discriminant Analysis	0.7040	0.7130	0.7222	0.5624	0.6192	0.3897	0.4101	0.0
ridge	Ridge Classifier	0.6960	0.7175	0.7222	0.5533	0.6139	0.3767	0.3974	0.0
et	Extra Trees Classifier	0.6960	0.7738	0.5306	0.6169	0.5332	0.3178	0.3435	0.0
ada	Ada Boost Classifier	0.6720	0.7266	0.5472	0.5681	0.5415	0.2938	0.3013	0.0
xgboost	Extreme Gradient Boosting	0.6720	0.7237	0.5056	0.5081	0.4895	0.2558	0.2666	0.0
lightgbm	Light Gradient Boosting Machine	0.6720	0.7097	0.4361	0.5038	0.4428	0.2253	0.2367	0.0
svm	SVM - Linear Kernel	0.6640	0.7317	0.6750	0.4991	0.5298	0.2943	0.3325	0.0
dummy	Dummy Classifier	0.6640	0.5000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0
nb	Naive Bayes	0.6560	0.6565	0.4500	0.5200	0.4647	0.2173	0.2267	0.0
knn	K Neighbors Classifier	0.6240	0.7275	0.6472	0.4683	0.5364	0.2413	0.2534	0.0

In [26]: `gbc_model = create_model('gbc')`

	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC
Fold							
0	0.8000	0.8897	0.6250	0.7143	0.6667	0.5247	0.5271
1	0.7200	0.7941	0.5000	0.5714	0.5333	0.3346	0.3361
2	0.8800	0.9559	0.8750	0.7778	0.8235	0.7331	0.7360
3	0.7200	0.7431	0.4444	0.6667	0.5333	0.3446	0.3590
4	0.6400	0.5486	0.4444	0.5000	0.4706	0.1993	0.2001
Mean	0.7520	0.7863	0.5778	0.6460	0.6055	0.4273	0.4317
Std	0.0816	0.1399	0.1626	0.0993	0.1264	0.1846	0.1843

```
In [27]: # Hyperparameter tuning

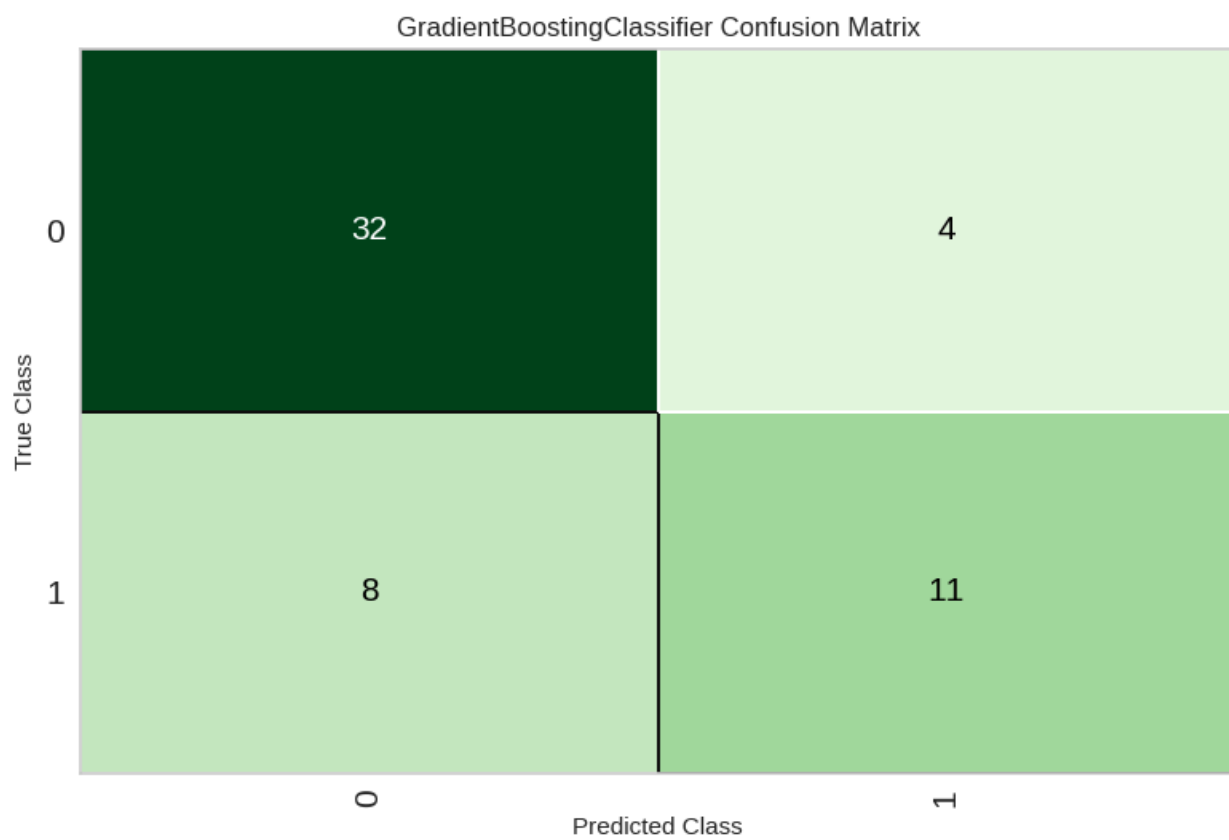
tuned_model1 = tune_model(gbc_model, optimize='AUC')
```

	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC
Fold							
0	0.8000	0.9191	0.7500	0.6667	0.7059	0.5552	0.5574
1	0.7200	0.8382	0.5000	0.5714	0.5333	0.3346	0.3361
2	0.8000	0.9191	0.8750	0.6364	0.7368	0.5819	0.6012
3	0.7200	0.8403	0.4444	0.6667	0.5333	0.3446	0.3590
4	0.6400	0.5347	0.3333	0.5000	0.4000	0.1573	0.1639
Mean	0.7360	0.8103	0.5806	0.6082	0.5819	0.3947	0.4035
Std	0.0599	0.1423	0.2008	0.0643	0.1242	0.1570	0.1592

Fitting 5 folds for each of 10 candidates, totalling 50 fits

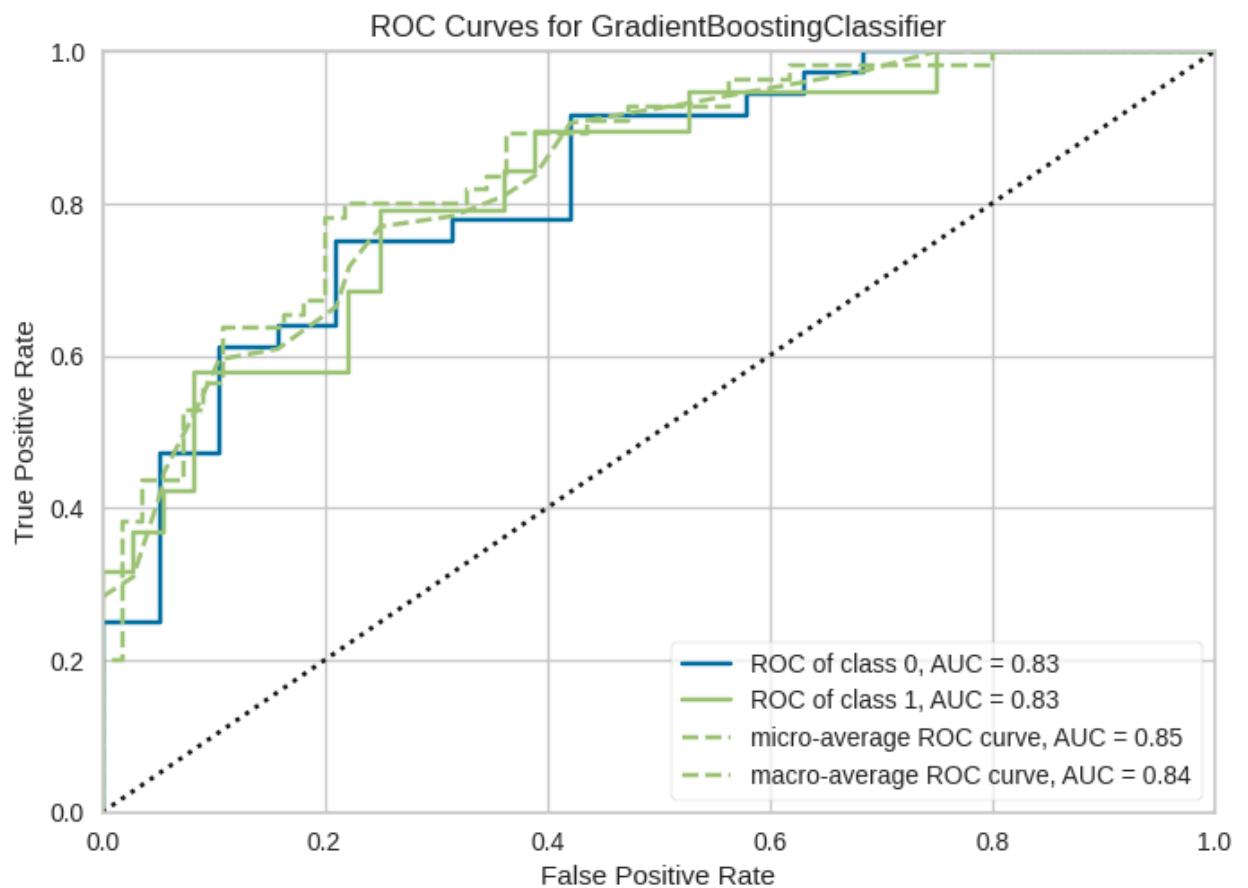
```
In [28]: # Confusion Matrix

plot_model(tuned_model1, plot='confusion_matrix')
```

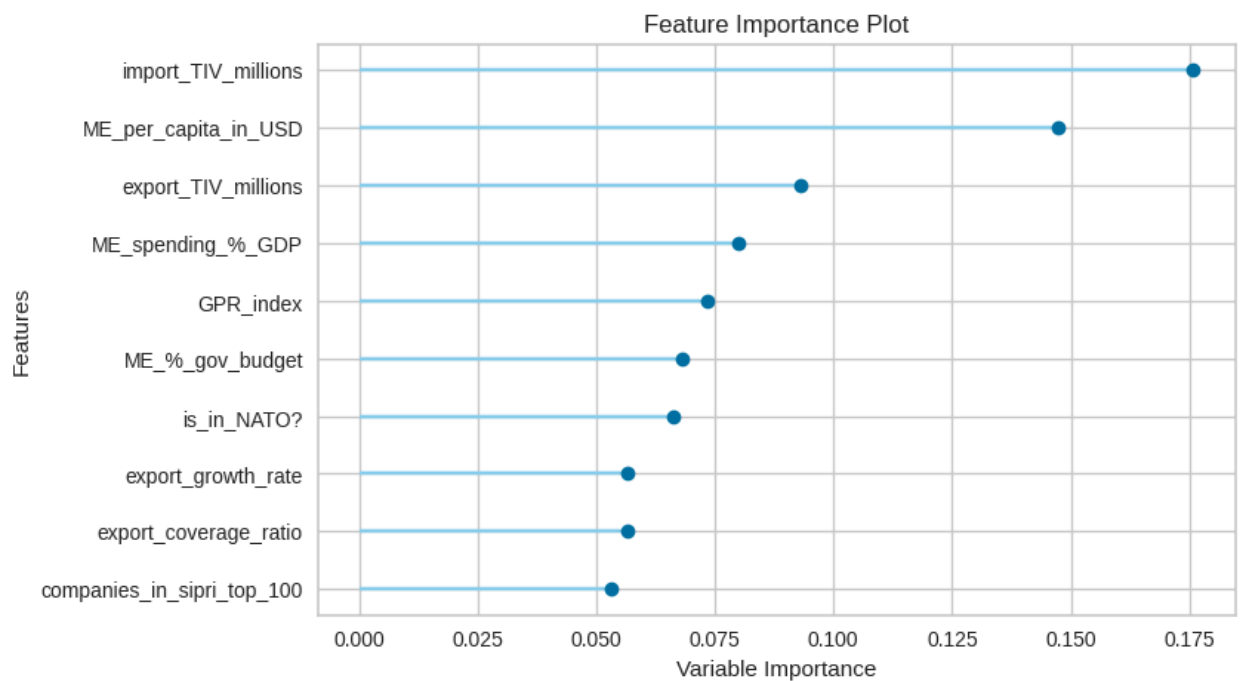


In [29]: *# ROC Curve*

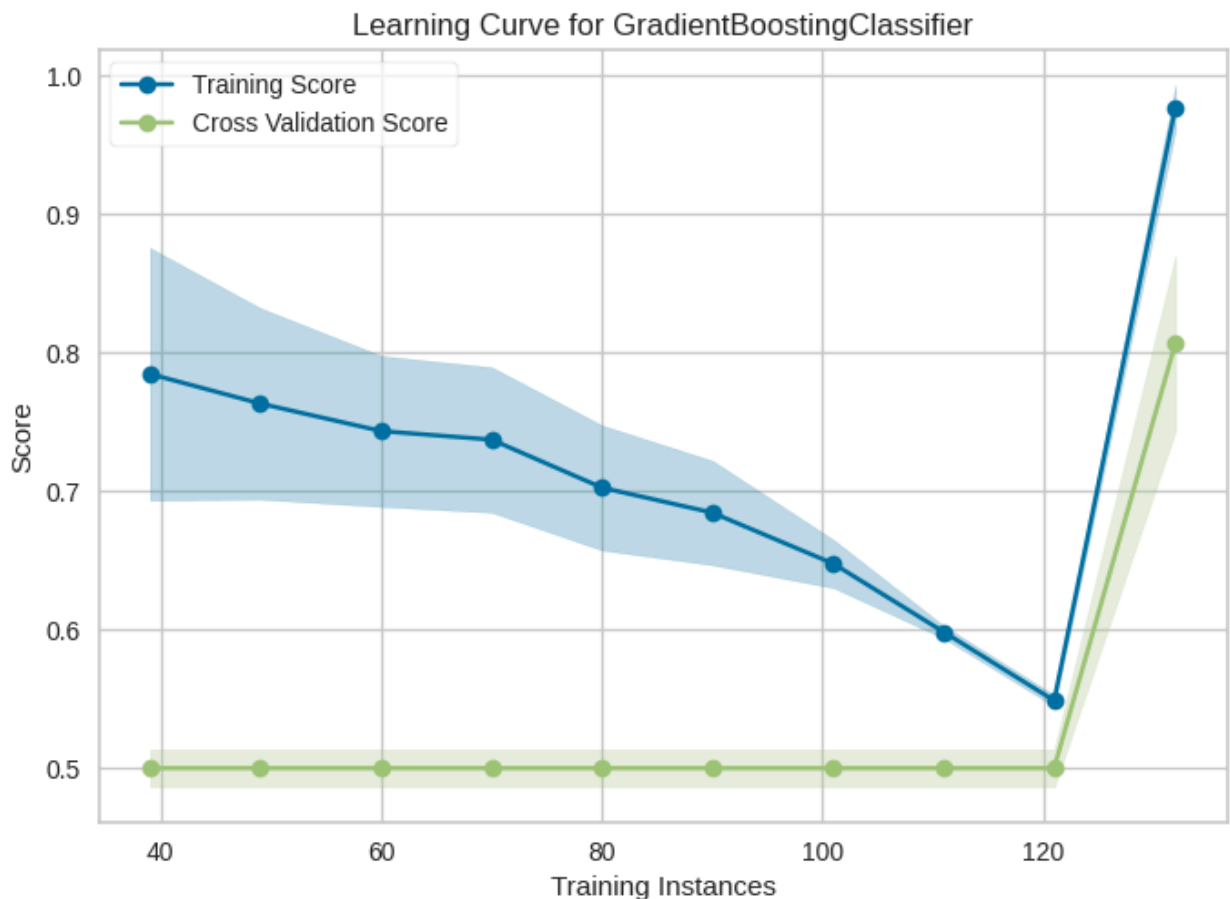
```
plot_model(tuned_model1, plot='auc')
```



```
In [30]: # Features that matter most
plot_model(tuned_model1, plot='feature')
```



```
In [31]: plot_model(tuned_model1, plot='learning')
```



The first attempt at building a classification model involves a binary classification as to whether the United States is a country's top arms supplier.

The best performing model is a Gradient Boosting Classifier. Our tuned Gradient Boosting Classifier held an accuracy of 0.74, an AUC of 0.81, a recall of 0.58, a precision of 0.61, and an F1 score of 0.58.

Our most important features are import TIV units (0.18) and military expenditure per capita (~0.14). We can interpret our Feature Importance Plot as showing that this was an interaction-heavy model.

8. Classification Modeling II: NATO Membership

```
In [32]: classification_pipeline2 = setup(
    data=df,
    target='is_in_NATO?',
    session_id=42,
    normalize=True,
    ignore_features=[
        'country',
        'year',
        'ME_spending_%_GDP_lead_1_year'
```

```
],  
    numeric_imputation='mean',  
    categorical_imputation='mode',  
    fix_imbalance=True,  
    fold=5  
)
```

	Description	Value
0	Session id	42
1	Target	is_in_NATO?
2	Target type	Binary
3	Original data shape	(180, 19)
4	Transformed data shape	(217, 38)
5	Transformed train set shape	(162, 38)
6	Transformed test set shape	(55, 38)
7	Ignore features	3
8	Numeric features	14
9	Categorical features	1
10	Rows with missing values	70.0%
11	Preprocess	True
12	Imputation type	simple
13	Numeric imputation	mean
14	Categorical imputation	mode
15	Maximum one-hot encoding	25
16	Encoding method	None
17	Fix imbalance	True
18	Fix imbalance method	SMOTE
19	Normalize	True
20	Normalize method	zscore
21	Fold Generator	StratifiedKFold
22	Fold Number	5
23	CPU Jobs	-1
24	Use GPU	False
25	Log Experiment	False
26	Experiment Name	clf-default-name
27	USI	8fbd

```
In [33]: # Compare all models
         best_model2 = compare_models()
```

	Model	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC	TF (S)
ada	Ada Boost Classifier	0.9040	0.9472	0.8875	0.9589	0.9189	0.8026	0.8115	0.0
gbc	Gradient Boosting Classifier	0.9040	0.9526	0.9132	0.9404	0.9247	0.7910	0.7961	0.0
xgboost	Extreme Gradient Boosting	0.8800	0.9277	0.8882	0.9229	0.9018	0.7463	0.7538	0.0
lightgbm	Light Gradient Boosting Machine	0.8800	0.9236	0.8875	0.9208	0.9036	0.7449	0.7463	0.0
rf	Random Forest Classifier	0.8720	0.9188	0.8647	0.9342	0.8966	0.7291	0.7361	0.0
et	Extra Trees Classifier	0.8720	0.8941	0.9125	0.8949	0.9006	0.7177	0.7269	0.0
dt	Decision Tree Classifier	0.8560	0.8528	0.8640	0.9091	0.8850	0.6923	0.6967	0.0
knn	K Neighbors Classifier	0.6800	0.7141	0.7412	0.7624	0.7500	0.3011	0.3039	0.0
lr	Logistic Regression	0.6720	0.7002	0.6897	0.7793	0.7261	0.3124	0.3213	0.0
ridge	Ridge Classifier	0.6640	0.6770	0.7022	0.7663	0.7241	0.2831	0.2956	0.0
lda	Linear Discriminant Analysis	0.6640	0.6785	0.7022	0.7663	0.7241	0.2831	0.2956	0.0
qda	Quadratic Discriminant Analysis	0.6480	0.7884	0.5779	0.8319	0.6737	0.3118	0.3421	0.0
svm	SVM - Linear Kernel	0.6080	0.6436	0.6529	0.7175	0.6822	0.1657	0.1687	0.0
nb	Naive Bayes	0.5120	0.7747	0.2949	0.8529	0.4351	0.1598	0.2332	0.0
dummy	Dummy Classifier	0.3520	0.5000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0

In [34]: `ada_model = create_model('ada')`

	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC
Fold							
0	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
1	0.9200	0.9653	0.9375	0.9375	0.9375	0.8264	0.8264
2	0.7600	0.8819	0.7500	0.8571	0.8000	0.5033	0.5104
3	0.8400	0.8889	0.7500	1.0000	0.8571	0.6835	0.7206
4	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Mean	0.9040	0.9472	0.8875	0.9589	0.9189	0.8026	0.8115
Std	0.0933	0.0521	0.1146	0.0564	0.0793	0.1909	0.1845

```
In [35]: # Hyperparameter tuning

tuned_model2 = tune_model(ada_model, optimize='AUC')
```

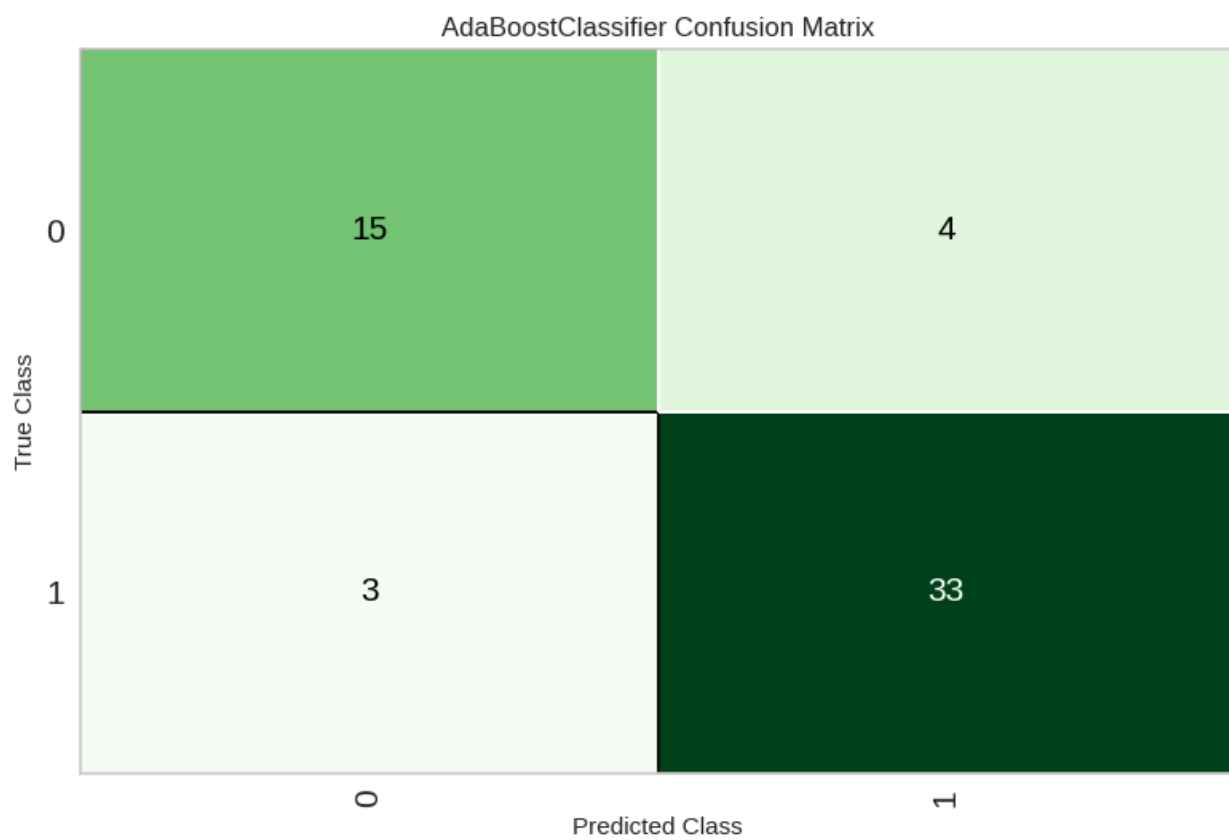
	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC
Fold							
0	0.9600	0.9861	1.0000	0.9412	0.9697	0.9110	0.9147
1	0.9200	0.9375	0.9375	0.9375	0.9375	0.8264	0.8264
2	0.8800	0.8542	0.9375	0.8824	0.9091	0.7331	0.7360
3	0.8000	0.8472	0.7500	0.9231	0.8276	0.5955	0.6138
4	0.9600	0.9926	1.0000	0.9444	0.9714	0.9049	0.9091
Mean	0.9040	0.9235	0.9250	0.9257	0.9231	0.7942	0.8000
Std	0.0599	0.0625	0.0919	0.0229	0.0530	0.1185	0.1136

Fitting 5 folds for each of 10 candidates, totalling 50 fits
 Original model was better than the tuned model, hence it will be returned. NOT
 E: The display metrics are for the tuned model (not the original one).

Pycaret's tune_model() returns our original model, as the tuned models did not perform better.

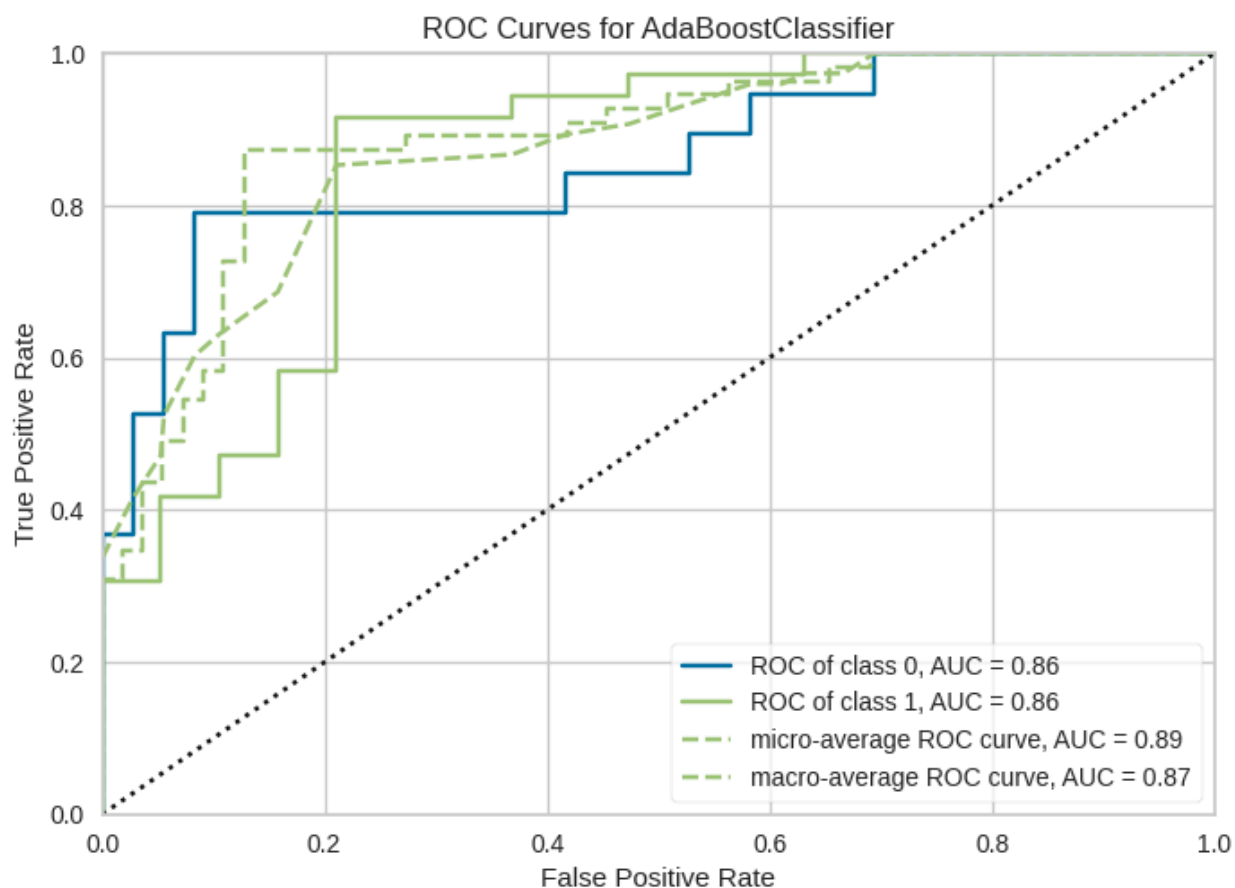
```
In [36]: # Confusion Matrix

plot_model(tuned_model2, plot='confusion_matrix')
```

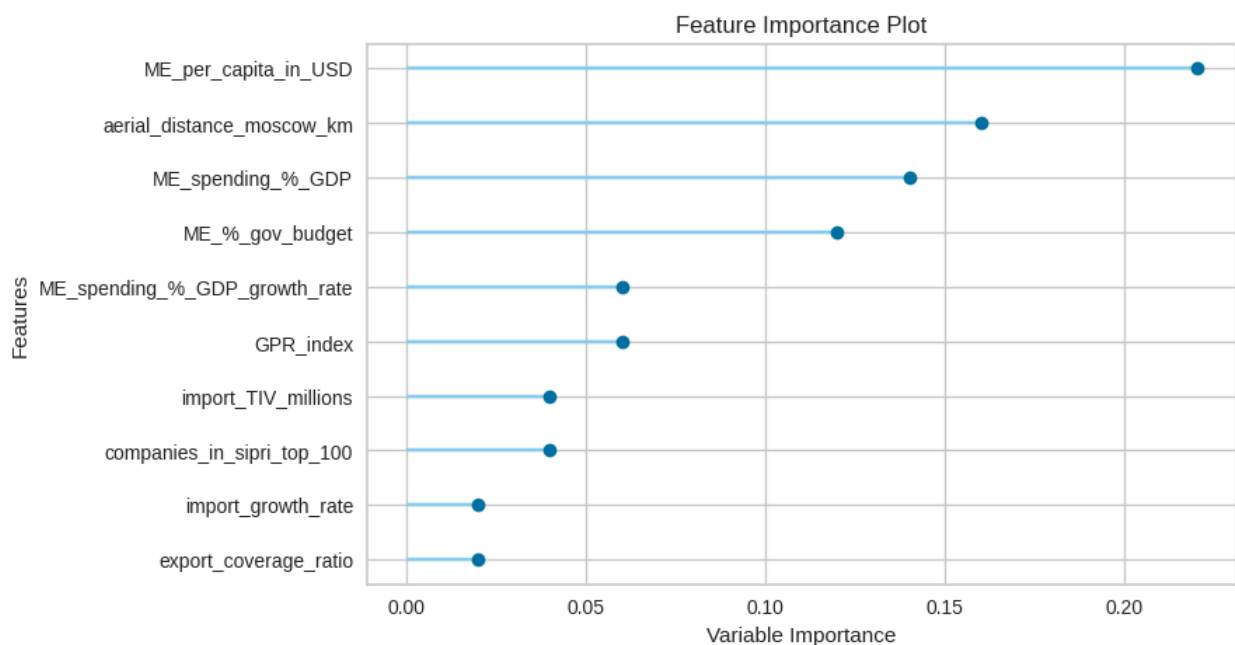


In [37]: *# ROC Curve*

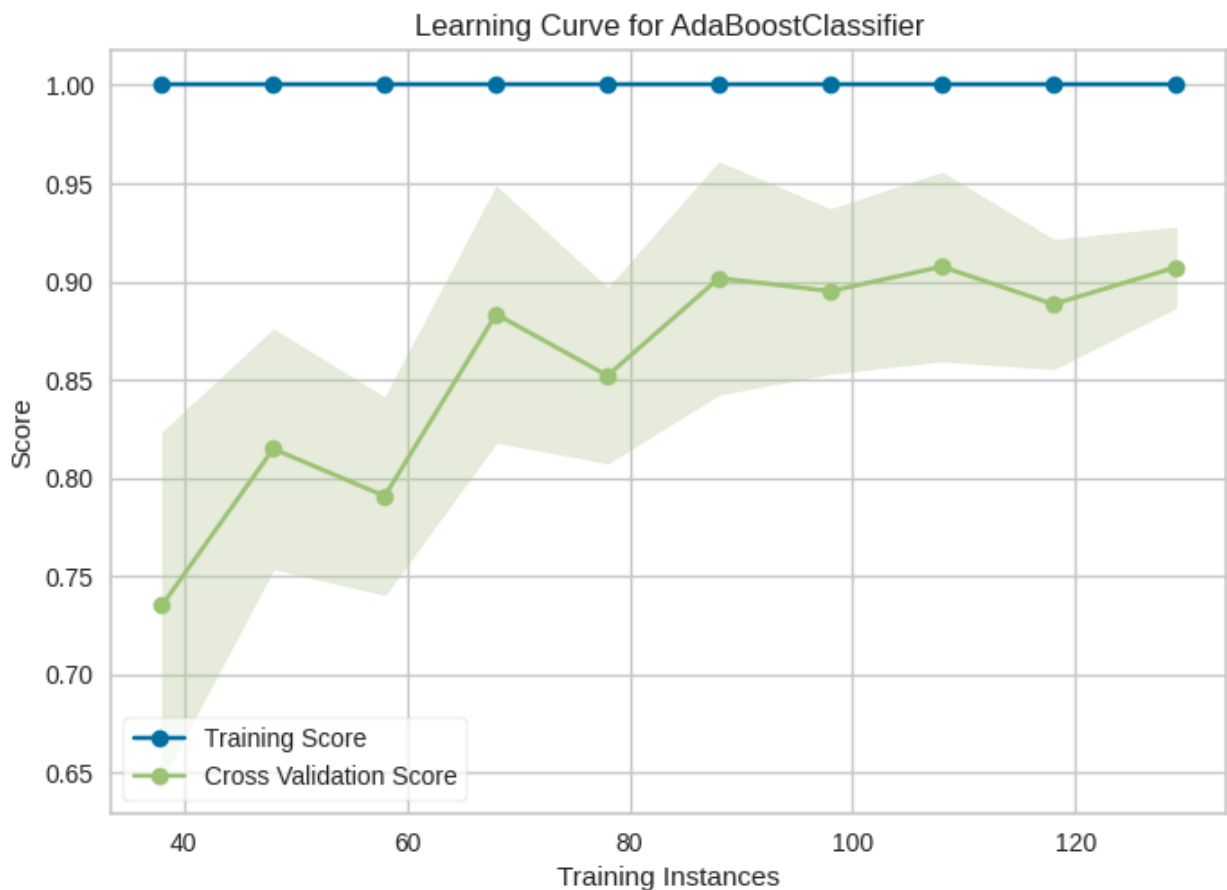
```
plot_model(tuned_model2, plot='auc')
```



```
In [38]: # Features that matter most
plot_model(tuned_model2, plot='feature')
```



```
In [39]: plot_model(tuned_model2, plot='learning')
```



The second attempt at building a classification model involves a binary classification as to whether a country is a member of NATO. One interesting caveat in this target data is that Finland became a NATO member in 2023, and Sweden a member in 2024.

The best performing model for this task was an AdaBoost Classifier. In contrast to other tree models like Extra Trees, AdaBoost builds decision trees in a sequence, having each new tree correct earlier classification mistakes. Our AdaBoost Classifier held an accuracy of 0.90, an AUC of 0.95, a recall of 0.89, a precision of 0.96, and an F1 score of 0.92. Across the board, this classifier performed better than our US Top Supplier Classifier on every metric. When using pycaret's `tune_model()` function, we didn't see any improvement over our original model.

Our most important features are Aerial Distance From Moscow (~0.16) and Military Expenditure Per Capita (~0.23). It makes sense that distance from Moscow would correlate highly with NATO membership, as NATO was a Western alliance that slowly moved east. Many countries closer to Moscow (Georgia, Ukraine, Moldova, etc.) are not members, as Russia has prevented their acceptance into the alliance by claiming territory within their borders. NATO members also have expected military spending targets, so it would make sense that a high military expenditure

per capita would correlate with NATO membership.

Decision Tree models usually come out on top when we pit several classification models against each other, but we can infer that the reason AdaBoost triumphed over other decision tree models was that the dataset had 2 very important features that together held a variable importance of around 0.39. With 2 features creating strong thresholds, AdaBoost's sequential tree design was uniquely suited to pick up on small additional observations missed that were not geographical (aerial distance from Moscow) or financial (military expenditure). To put it simply, all decision tree models picked up on *aerial_distance_moscow_km* and *ME_per_capita_in_USD* being the most important features, but AdaBoost was able to find minor tweaks for other features once it understood the major importance of the first two.

9. Clustering and Dimensionality Reduction: Finding Commonalities

```
In [40]: from pycaret.clustering import *
```

```
In [41]: from scipy.cluster.hierarchy import dendrogram, linkage

# This function helps create all future dendrogram visualizations.

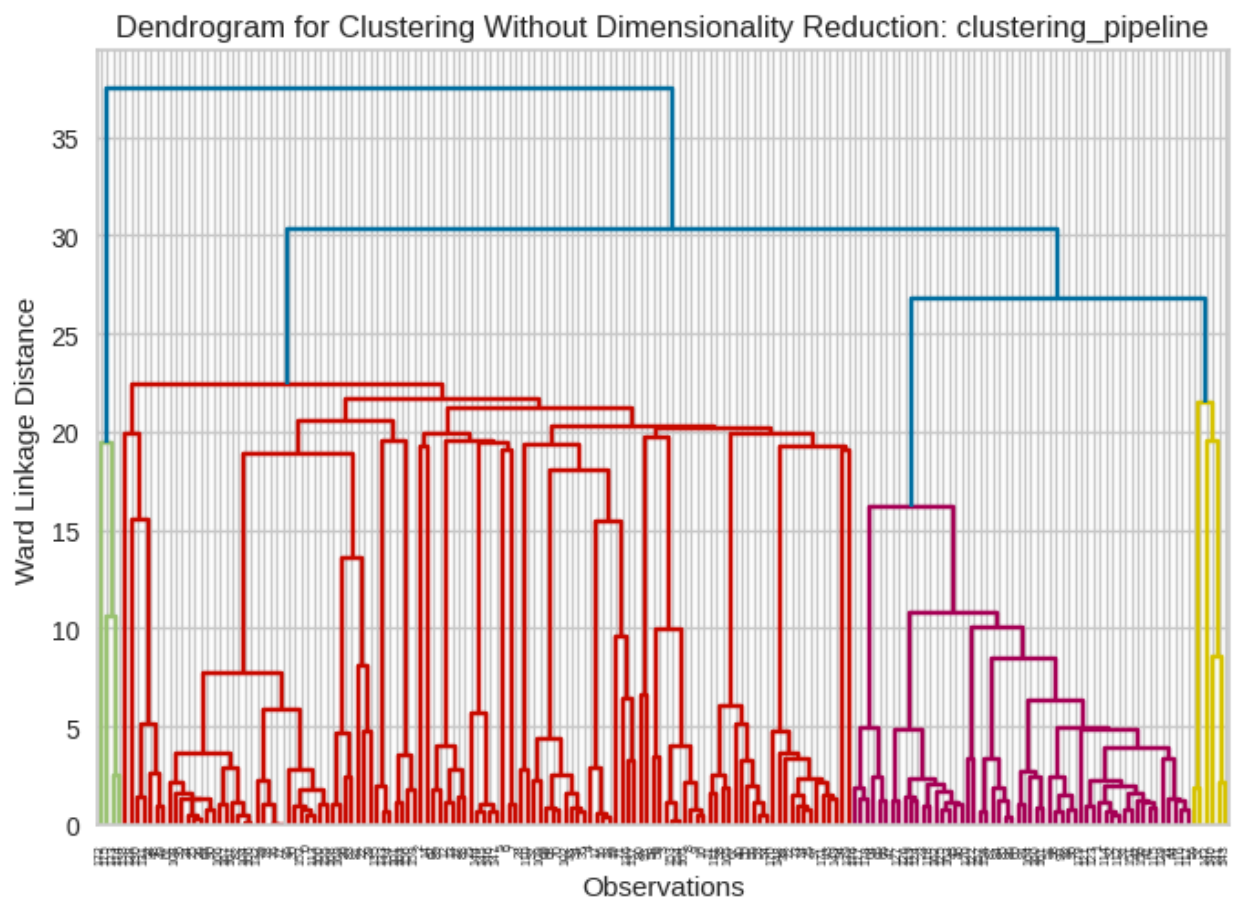
def dendro(title):
    dendrogram(linkage(get_config('X_transformed'), method='ward'))
    plt.xlabel('Observations')
    plt.ylabel('Ward Linkage Distance')
    plt.title(title)
    plt.show()
```

9.1 Clustering Without Dimensionality Reduction

```
In [42]: clustering_pipeline = setup(
    data=df,
    ignore_features=[
        'country',
        'year'
    ],
    numeric_imputation='mean',
    categorical_imputation='mode',
    normalize=True,
    normalize_method='zscore',
    session_id=42
)
```

	Description	Value
0	Session id	42
1	Original data shape	(180, 19)
2	Transformed data shape	(180, 41)
3	Ignore features	2
4	Numeric features	16
5	Categorical features	1
6	Rows with missing values	70.0%
7	Preprocess	True
8	Imputation type	simple
9	Numeric imputation	mean
10	Categorical imputation	mode
11	Maximum one-hot encoding	-1
12	Encoding method	None
13	Normalize	True
14	Normalize method	zscore
15	CPU Jobs	-1
16	Use GPU	False
17	Log Experiment	False
18	Experiment Name	cluster-default-name
19	USI	0294

```
In [43]: dendro("Dendrogram for Clustering Without Dimensionality "
               "Reduction: clustering_pipeline")
```



This looks atrocious compared to the results we'll get when we use dimensionality reduction.

```
In [44]: print(models())
```

ID	Name \
kmeans	K-Means Clustering
ap	Affinity Propagation
meanshift	Mean Shift Clustering
sc	Spectral Clustering
hclust	Agglomerative Clustering
dbscan	Density-Based Spatial Clustering
optics	OPTICS Clustering
birch	Birch Clustering

ID	Reference
kmeans	sklearn.cluster._kmeans.KMeans
ap	sklearn.cluster._affinity_propagation.Affinity...
meanshift	sklearn.cluster._mean_shift.MeanShift
sc	sklearn.cluster._spectral.SpectralClustering
hclust	sklearn.cluster._agglomerative.AgglomerativeCl...
dbscan	sklearn.cluster._dbscan.DBSCAN
optics	sklearn.cluster._optics.OPTICS
birch	sklearn.cluster._birch.Birch

```
In [45]: kmeans_model = create_model('kmeans', num_clusters=4)
```

	Silhouette	Calinski-Harabasz	Davies-Bouldin	Homogeneity	Rand Index	Completeness
0	0.2167	11.6329	1.1496	0	0	0

```
In [46]: hclust_model = create_model('hclust', num_clusters=6)
```

	Silhouette	Calinski-Harabasz	Davies-Bouldin	Homogeneity	Rand Index	Completeness
0	0.1231	13.0067	2.3107	0	0	0

The first attempt at clustering does not involve dimensionality reduction. It produced results that were not significant or meaningful. The dendrogram plot is unreadable, and we're left with our best silhouette score being only 0.22 (a k-means clustering model).

9.2 Linear PCA + Clustering

```
In [47]: clustering_pipeline2 = setup(
    data=df,
    ignore_features=[
        'country',
        'year'
    ],
    numeric_imputation='mean',
    categorical_imputation='mode',
```



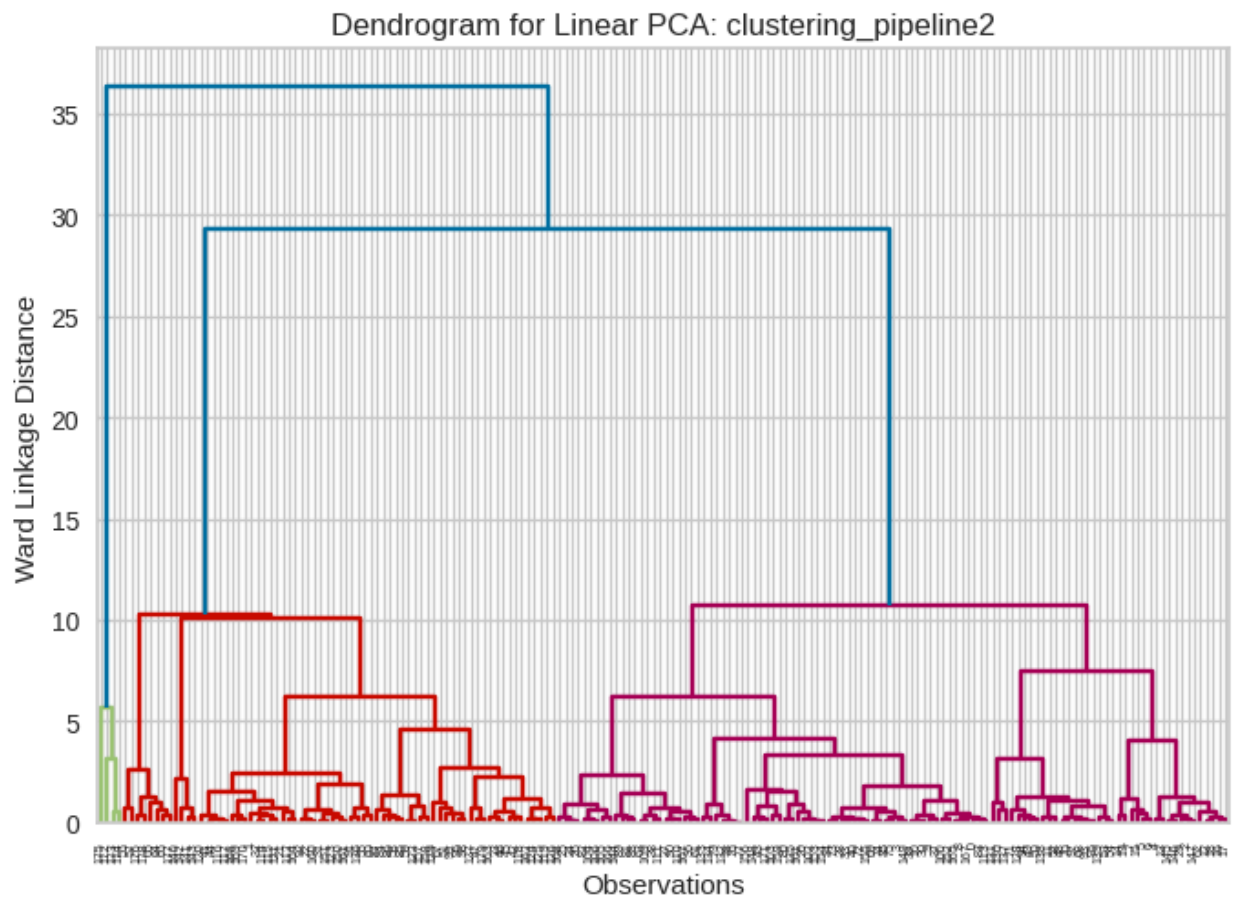
```

normalize=True,
normalize_method='zscore',
session_id=42,
pca=True,
pca_method='linear',
pca_components=2
)

```

	Description	Value
0	Session id	42
1	Original data shape	(180, 19)
2	Transformed data shape	(180, 2)
3	Ignore features	2
4	Numeric features	16
5	Categorical features	1
6	Rows with missing values	70.0%
7	Preprocess	True
8	Imputation type	simple
9	Numeric imputation	mean
10	Categorical imputation	mode
11	Maximum one-hot encoding	-1
12	Encoding method	None
13	Normalize	True
14	Normalize method	zscore
15	PCA	True
16	PCA method	linear
17	PCA components	2
18	CPU Jobs	-1
19	Use GPU	False
20	Log Experiment	False
21	Experiment Name	cluster-default-name
22	USI	8cdc

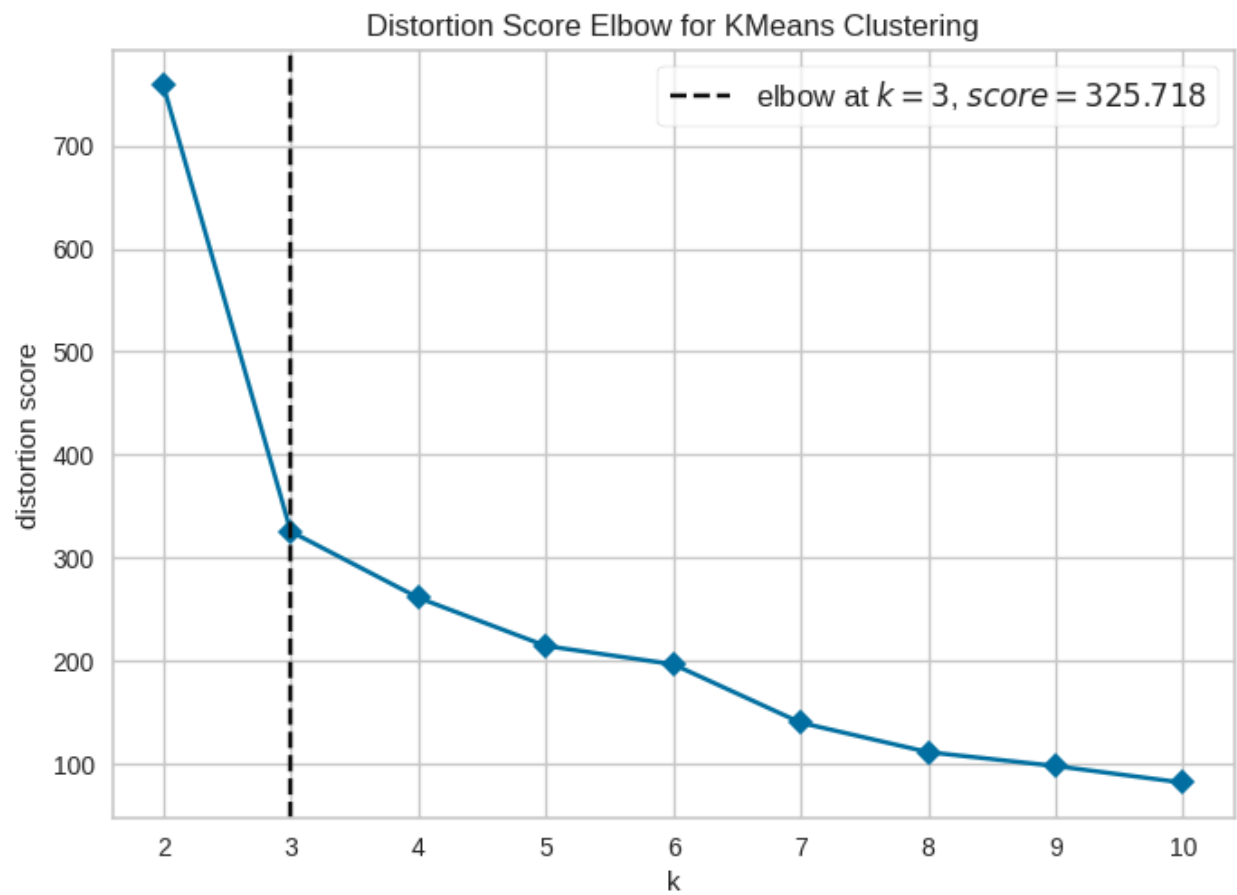
```
In [48]: dendro("Dendrogram for Linear PCA: clustering_pipeline2")
```



```
In [49]: kmeans_model2 = create_model('kmeans', num_clusters=3)
```

	Silhouette	Calinski-Harabasz	Davies-Bouldin	Homogeneity	Rand Index	Completeness
0	0.5384	297.2208	0.5493	0	0	0

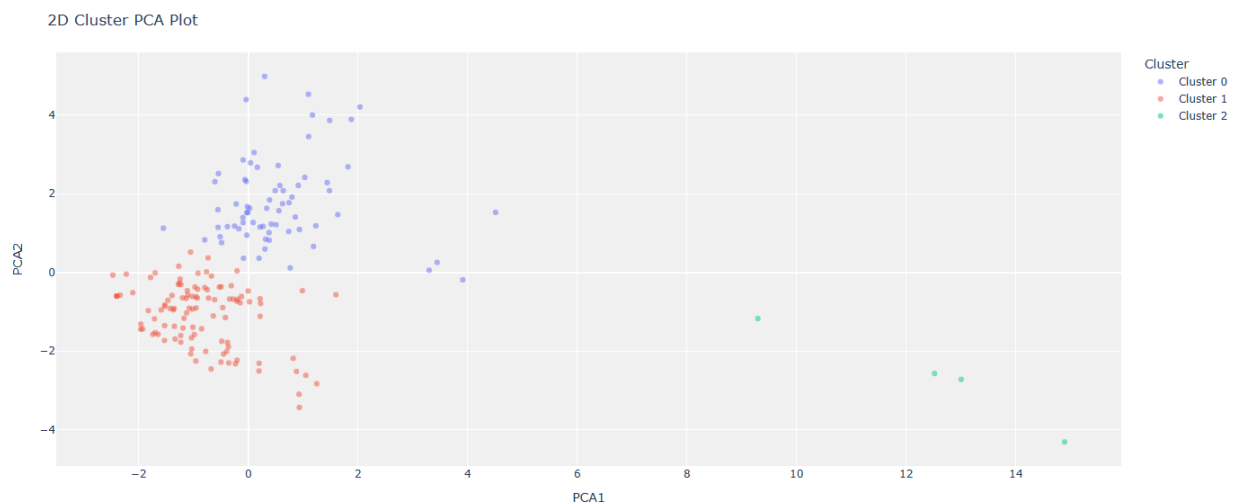
```
In [50]: plot_model(kmeans_model2, plot='elbow')
```



```
In [51]: plot_model(kmeans_model2, plot='cluster')
```

In [52]: *# The above cluster plot has trouble rendering correctly in most notebooks.
Here's a .png file of what it looks like.*

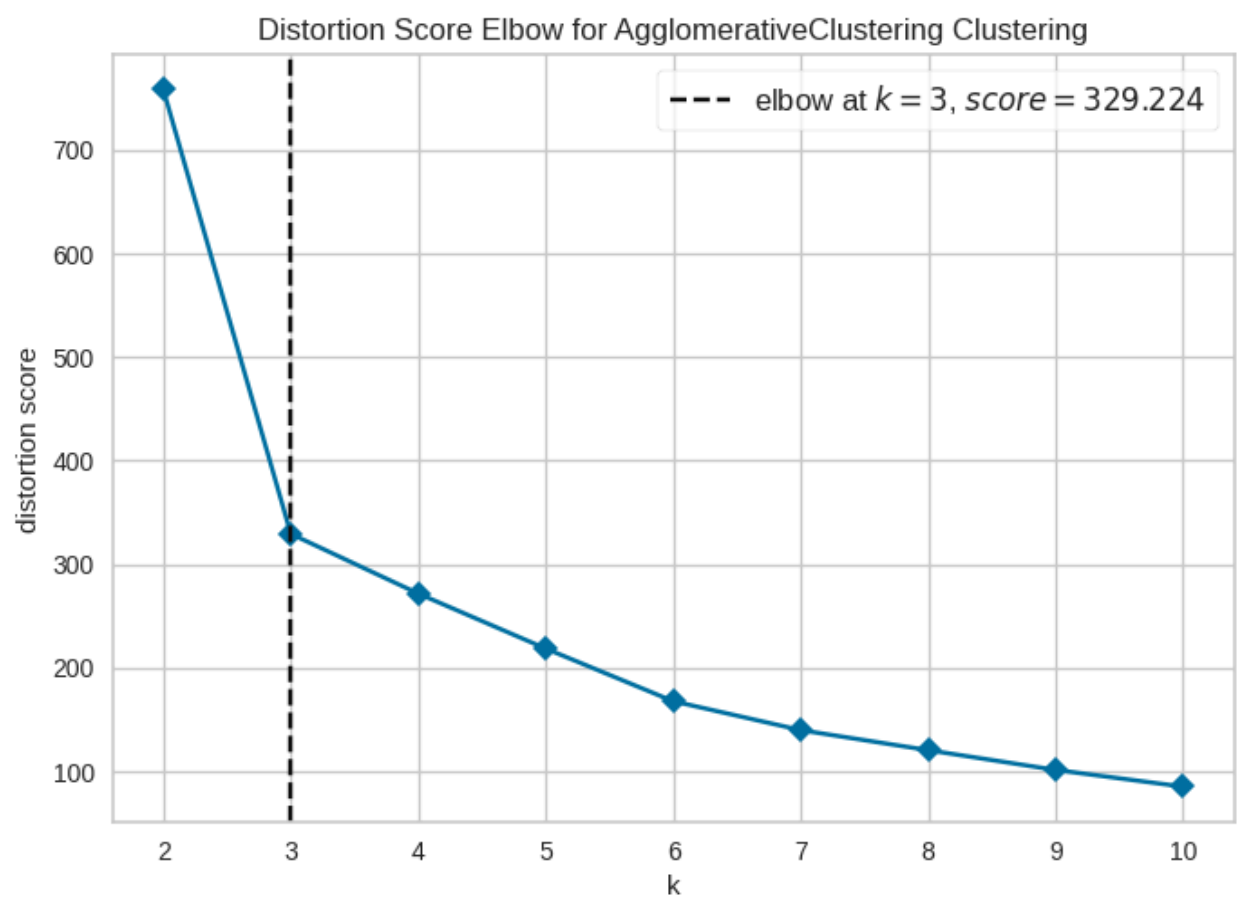
```
display(Image(filename="linear_PCA_kmeans.png", width=1100))
```



```
In [53]: hclust_model2 = create_model('hclust', num_clusters=3)
```

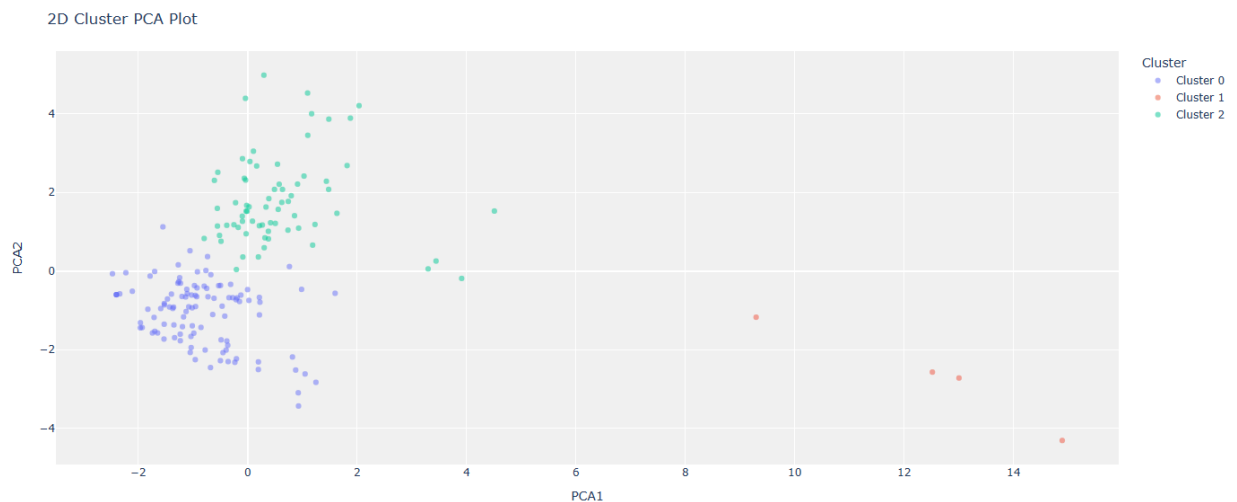
	Silhouette	Calinski-Harabasz	Davies-Bouldin	Homogeneity	Rand Index	Completeness
0	0.5327	293.1130	0.5519	0	0	0

```
In [54]: plot_model(hclust_model2, plot='elbow')
```



```
In [55]: plot_model(hclust_model2, plot='cluster')
```

```
In [56]: # The above cluster plot has trouble rendering correctly in most notebooks.  
# Here's a .png file of what it looks like.  
  
display(Image(filename="linear_PCA_hclust.png", width=1100))
```



The second clustering attempt uses linear PCA as its method of dimensionality reduction with improved results. When reducing our features down to two linear principal components, we get a dendrogram and distortion score elbow plot that favors 3 clusters (in both our k-means clustering model and our hierarchical agglomerative clustering model). When creating 2D scatter plots of the clusters, our k-means model and our hierarchical agglomerative model give us nearly identical results, with the only difference being small border data points between the top-left corner and bottom-left corner clusters. The top-left corner cluster groups together countries known to be major military import or export players: Russia, France, Turkey, United Kingdom, Poland, Italy, and Germany. The bottom-left corner cluster groups together countries that have relatively low military spending needs: Ireland, Iceland, Malta, Kosovo, etc. Ukraine, (a strong outlier as an active war zone,) becomes its own cluster. The k-means model gives us the slightly better of the two silhouette scores at 0.54.

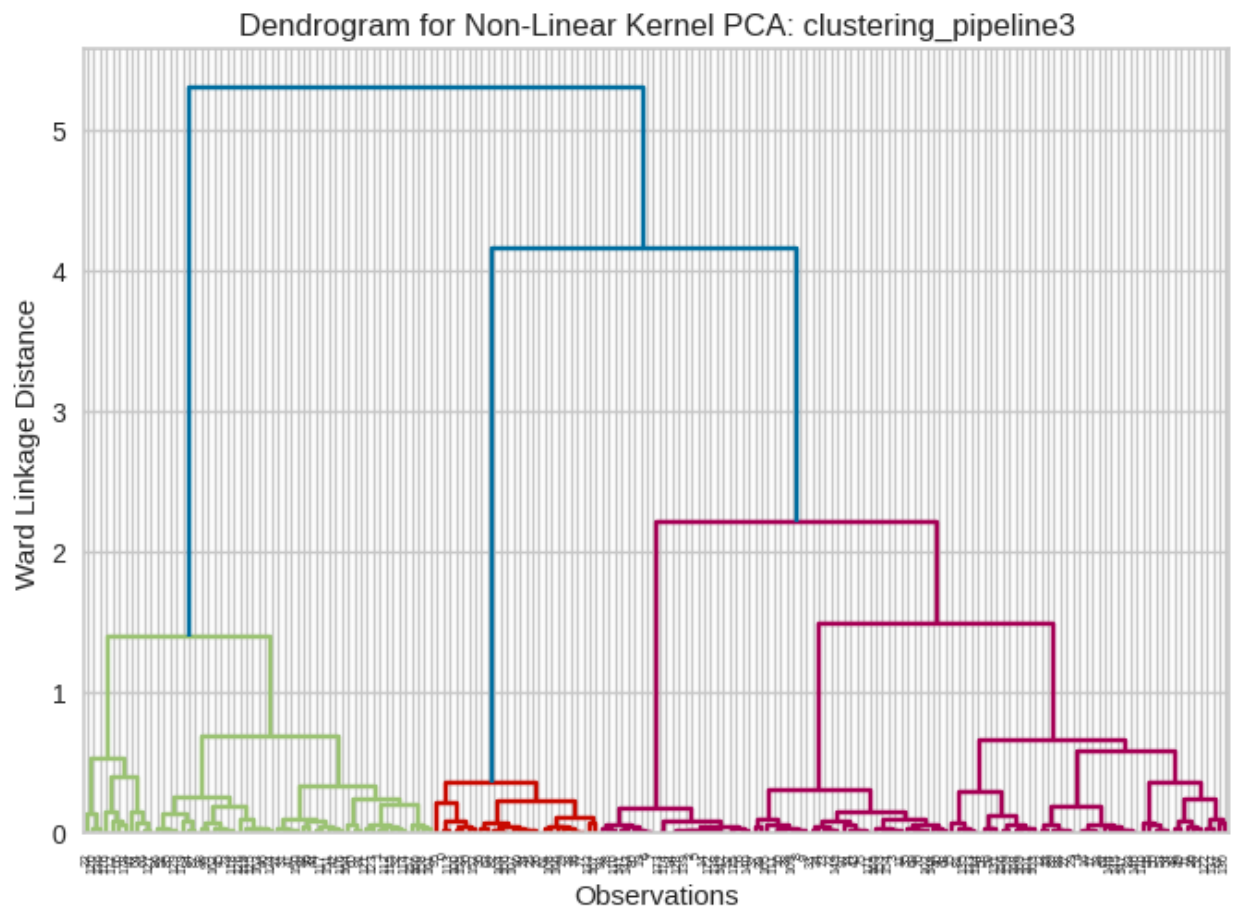
9.3 Nonlinear Kernel PCA + Clustering

```
In [57]: clustering_pipeline3 = setup(
    data=df,
    ignore_features=[
        'country',
        'year'
    ],
    numeric_imputation='mean',
    categorical_imputation='mode',
    normalize=True,
    normalize_method='zscore',
    session_id=42,
    pca=True,
    pca_method='kernel',
    pca_components=2
```

)

	Description	Value
0	Session id	42
1	Original data shape	(180, 19)
2	Transformed data shape	(180, 2)
3	Ignore features	2
4	Numeric features	16
5	Categorical features	1
6	Rows with missing values	70.0%
7	Preprocess	True
8	Imputation type	simple
9	Numeric imputation	mean
10	Categorical imputation	mode
11	Maximum one-hot encoding	-1
12	Encoding method	None
13	Normalize	True
14	Normalize method	zscore
15	PCA	True
16	PCA method	kernel
17	PCA components	2
18	CPU Jobs	-1
19	Use GPU	False
20	Log Experiment	False
21	Experiment Name	cluster-default-name
22	USI	314a

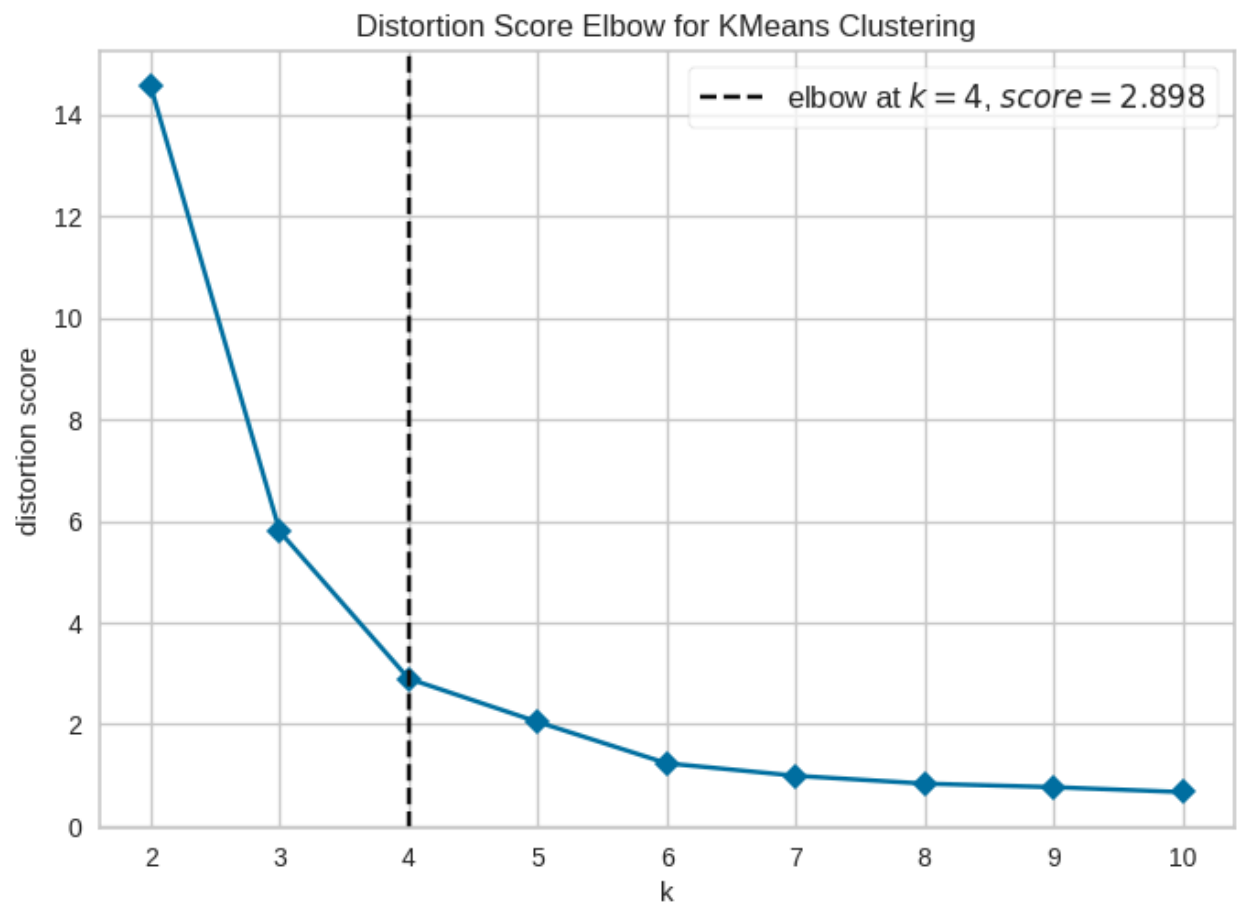
```
In [58]: dendro("Dendrogram for Non-Linear Kernel PCA: clustering_pipeline3")
```

```
In [59]: kmeans_model3 = create_model('kmeans', num_clusters=3)
```

	Silhouette	Calinski-Harabasz	Davies-Bouldin	Homogeneity	Rand Index	Completeness
0	0.6504	349.3418	0.4634	0	0	0

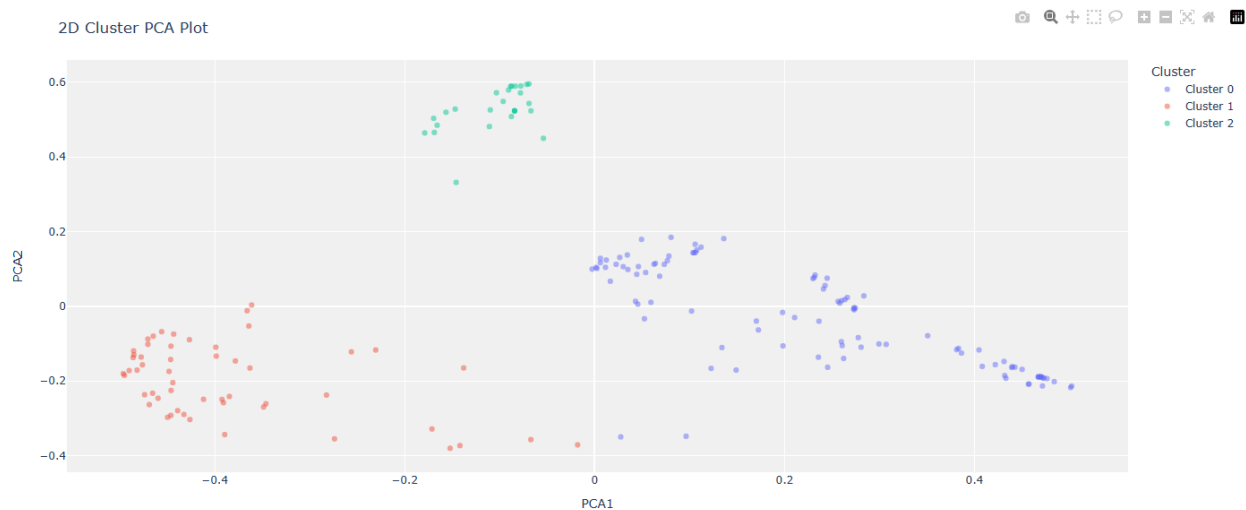
```
In [60]: plot_model(kmeans_model3, plot='elbow')
```



```
In [61]: plot_model(kmeans_model3, plot='cluster')
```

In [62]: *# The above cluster plot has trouble rendering correctly in most notebooks.
Here's a .png file of what it looks like.*

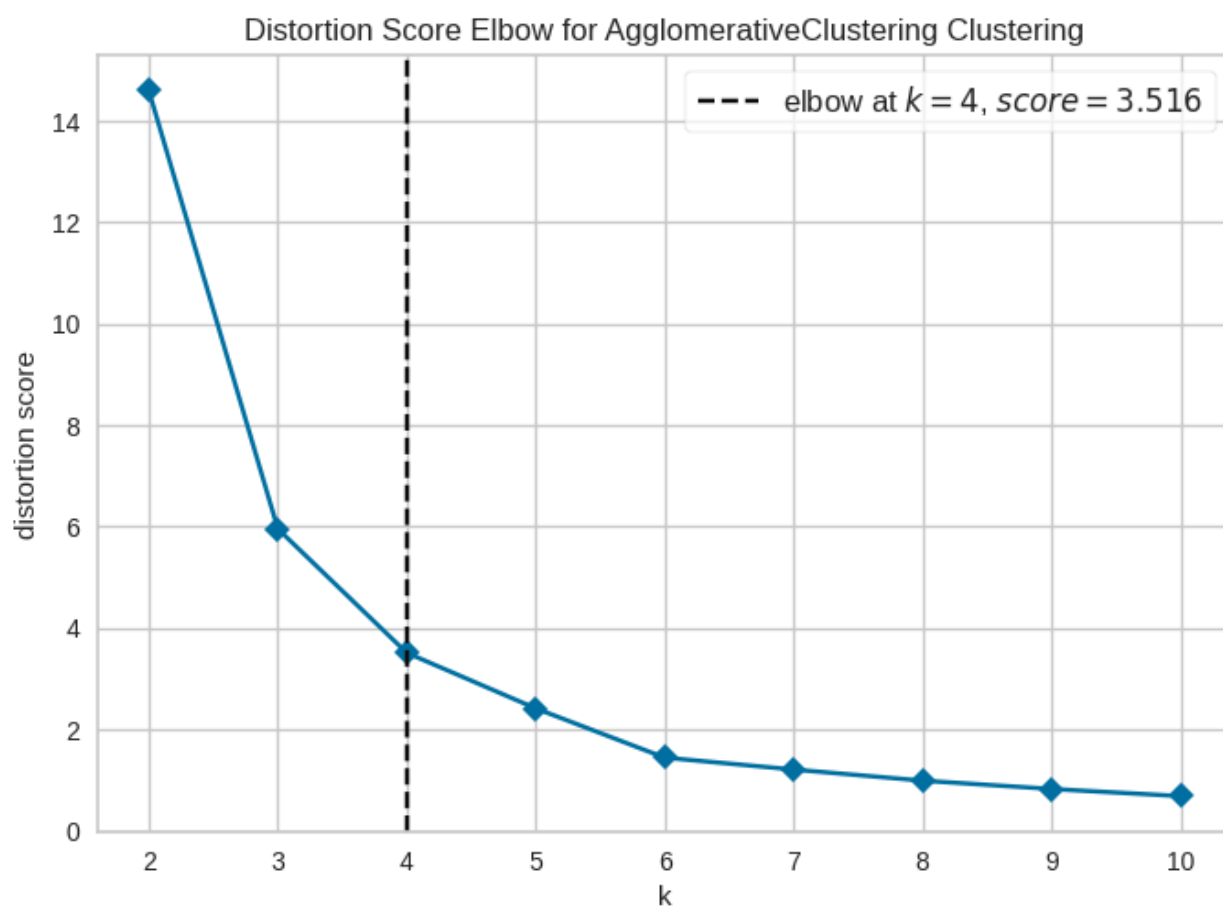
```
display(Image(filename="nonlinear_PCA_kmeans.png", width=1100))
```



```
In [63]: hclust_model3 = create_model('hclust', num_clusters=3)
```

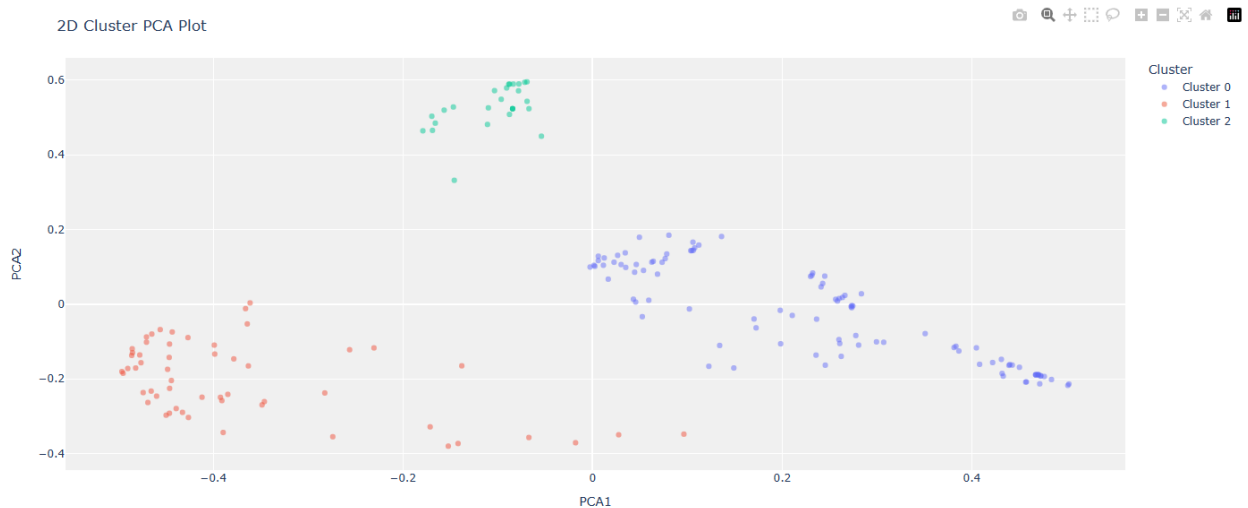
	Silhouette	Calinski-Harabasz	Davies-Bouldin	Homogeneity	Rand Index	Completeness
0	0.6450	337.0767	0.4802	0	0	0

```
In [64]: plot_model(hclust_model3, plot='elbow')
```



```
In [65]: plot_model(hclust_model3, plot='cluster')
```

```
In [66]: # The above cluster plot has trouble rendering correctly in most notebooks.  
# Here's a .png file of what it looks like.  
  
display(Image(filename="nonlinear_PCA_hclust.png", width=1100))
```



The third clustering attempt uses nonlinear kernel PCA as its method of dimensionality reduction. When reducing our features down to two non-linear principal components, we get a dendrogram that favors 3 clusters (in both our k-means model and our hierarchical agglomerative model). When creating 2D scatter plots of the clusters, our k-means model and our agglomerative model once again give us nearly identical results (the main difference being a few United Kingdom data points at the bottom center of both plots).

The top-center of the scatter plot gives us our tightest cluster, composed mostly of smaller countries without much need in terms of military spending: Iceland, Portugal, Malta, Montenegro, etc. Many of these countries have periods of low or no arms imports. The bottom-right cluster includes countries with major military capabilities, (Russia, Ukraine, France, Poland, Turkey) but many additional countries that don't form clear similarities. In the bottom-left corner, we get a cluster of many NATO members: Norway, Sweden, Lithuania, Netherlands, Slovakia, Germany, etc. Another commonality of this cluster is that most of the countries have the US as their top arms supplier. Both the k-means model and the agglomerative model give us our best resulting silhouette score: 0.65.

It should be noted that Ukraine's data points were a massive outlier. When forcing the number of clusters to be exactly 2, all 4 of the plotted models preferred clustering Ukraine on its own, with the remaining cluster containing all other European countries. Ukraine is the main warfront, having transformed its economy into an engine to sustain a long-term, grinding war of attrition.

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